Section: Module 7

Sensitivity Analysis

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Agenda

- Observational Studies
- Sensitivity Analysis
  1. Partial R^2 / Omitted Variable Bias
  2. Cornfield Condition
     - We can quantify the robustness using these two approaches (robustness value / E-value).
  3. Rosenbaum's Γ (next week)
  4. Marginal Sensitivity Model (next week)
- Partial identification
- Review Questions
  - Regression: omitted-variable bias, standard errors, and selection
  - Sensitivity analysis: LaLonde job-training application (Heckman selection model)
- Modeling Selection Bias with Heckman Selection model (Appendix)

Identification of the Average Treatment Effect (1)

- Our quantity of interest is

$$\tau = \mathbb{E}\{Y_i(1) - Y_i(0)\}.$$

- Consistency: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$.
- Unconfoundedness:

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp T_i \mid X_i.$$

- Overlap: for every x in the support of X_i,

$$0 < \Pr(T_i = 1 \mid X_i = x) < 1.$$

Identification of the Average Treatment Effect (2)

- Let $\mu_t(x) = \mathbb{E}[Y_i \mid T_i = t, X_i = x]$.
- For each $t \in \{0, 1\}$, we have

$$\begin{aligned}\mathbb{E}[Y_i(t) \mid X_i = x] &= \mathbb{E}[Y_i(t) \mid T_i = t, X_i = x] \quad \because \text{(Unconfoundedness)} \\ &= \mathbb{E}[Y_i \mid T_i = t, X_i = x] \quad (\because \text{Consistency}) \\ &= \mu_t(x).\end{aligned}$$

- By iterated expectations and the previous result,

$$\tau = \mathbb{E}\big[\mathbb{E}\{Y_i(1) - Y_i(0) \mid X_i\}\big] = \mathbb{E}\big[\mu_1(X_i) - \mu_0(X_i)\big].$$

- The identification result suggests a plug-in estimator.
  1. Estimate $\mu_1(x)$ and $\mu_0(x)$.
  2. Predict both outcomes for every respondent: $\widehat{\mu}_1(X_i)$ and $\widehat{\mu}_0(X_i)$.
  3. Average their difference:

$$\widehat{\tau}_{\text{reg}} = \frac{1}{n}\sum_{i=1}^{n}\{\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)\}.$$

ATE from Linear Regression

- Suppose the conditional mean is additive:

$$\mu_t(x) = \alpha + \beta t + x^\top \gamma.$$

  Then $\mu_1(x) - \mu_0(x) = \beta$, so $\widehat{\tau}_{\text{reg}} = \widehat{\beta}$.

- With interactions between treatment and covariates,

$$\mu_t(x) = \alpha + \beta t + x^\top \gamma + t x^\top \delta.$$

  Thus,

$$\mu_1(x) - \mu_0(x) = \beta + x^\top \delta,$$

$$\widehat{\tau}_{\text{reg}} = \widehat{\beta} + \overline{X}^\top \widehat{\delta}, \qquad \overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

ATE from Nonlinear Regression

- For a binary outcome, consider a logistic regression:

$$\mu_t(x) = \Lambda(\alpha + \beta t + x^\top \gamma), \qquad \Lambda(a) = \frac{\exp(a)}{1 + \exp(a)}.$$

- First, predict the outcome under each treatment value:

$$\widehat{\mu}_t(X_i) = \Lambda(\widehat{\alpha} + \widehat{\beta} t + X_i^\top \widehat{\gamma}).$$

- Then, average the differences:

$$\widehat{\tau}_{\text{reg}} = \frac{1}{n} \sum_{i=1}^{n} \{\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)\}.$$

- We can obtain its variance using the delta method.

Sensitivity Analysis (1)

- So far, identification relies on unconfoundedness given the observed covariates.
- Suppose we also need an unobserved confounder U_i:

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp T_i \mid \boldsymbol{X}_i, U_i.$$

  Controlling for $\boldsymbol{X}_i$ alone may leave confounding.

- **Sensitivity analysis**: How strong would the associations of U_i with treatment and outcome have to be to change our conclusion?
- We vary assumptions about these associations and examine the resulting coefficient or causal-effect bound.

Sensitivity Analysis (2)

- There are many approaches to sensitivity analysis
  1. **Partial R^2 / omitted variable bias** (Cinelli and Hazlett, 2020, JRSS-B)
     - Express the omitted-variable bias using the residual variation in treatment and outcome explained by U_i.
  2. **Cornfield condition / E-value** (Ding and VanderWeele, 2016, Epidemiology)
     - Bound the causal risk ratio using the strengths of the treatment–confounder and confounder–outcome associations.
  3. **Rosenbaum's Γ** (Rosenbaum, 2002, Observational Studies)
  4. **Marginal sensitivity model** (Tan, 2006, Biometrika)

- Other approaches: modeling selection bias
  - Heckman selection model (Heckman, 1979, Econometrica)
  - Control function approach

- Today, we go over (1) and (2) (partial R^2 and Cornfield condition).

Omitted Variable Bias Formula (1)

- Suppose that the true model is

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \delta U_i + \epsilon_i, \qquad \mathbb{E}[\epsilon_i \mid T_i, \boldsymbol{X}_i, U_i] = 0,$$

  but you use the model

$$Y_i = \alpha^* + \beta^* T_i + \boldsymbol{\gamma}^{*\top} \boldsymbol{X}_i + Y_i^{\perp T, \boldsymbol{X}}.$$

  Here $Y_i^{\perp T, \boldsymbol{X}}$ is the residual from this regression.
- Recall that the FWL theorem tells us

$$\beta^* = \frac{\mathrm{Cov}(Y_i, \tilde{T}_i^*)}{\mathbb{V}[\tilde{T}_i^*]},$$

  where

$$T_i = \phi_0^* + \boldsymbol{\phi}_1^{*\top} \boldsymbol{X}_i + \tilde{T}_i^*.$$

  All regressions include an intercept.

Omitted Variable Bias Formula (2)

- Then,

$$\begin{aligned}\beta^* &= \frac{\mathrm{Cov}(\alpha + \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \delta U_i + \epsilon_i, \tilde{T}_i^*)}{\mathbb{V}[\tilde{T}_i^*]} \\ &= \frac{\mathrm{Cov}(\beta T_i + \delta U_i, \tilde{T}_i^*)}{\mathbb{V}[\tilde{T}_i^*]} \\ &= \beta + \delta \times \frac{\mathrm{Cov}(U_i, \tilde{T}_i^*)}{\mathbb{V}[\tilde{T}_i^*]}.\end{aligned}$$

  The terms involving $\boldsymbol{X}_i$ and ϵ_i vanish by orthogonality. The last line uses

$$\mathrm{Cov}(T_i, \tilde{T}_i^*) = \mathrm{Cov}(\phi_0^* + \boldsymbol{\phi}_1^{*\top} \boldsymbol{X}_i + \tilde{T}_i^*, \tilde{T}_i^*) = \mathbb{V}[\tilde{T}_i^*].$$

- Also, consider $U_i = \psi_0^* + \boldsymbol{\psi}_1^\top \boldsymbol{X}_i + \tilde{U}_i$. Then,

$$\mathrm{Cov}(U_i, \tilde{T}_i^*) = \mathrm{Cov}(\psi_0^* + \boldsymbol{\psi}_1^\top \boldsymbol{X}_i + \tilde{U}_i, \tilde{T}_i^*) = \mathrm{Cov}(\tilde{U}_i, \tilde{T}_i^*).$$

Omitted Variable Bias Formula (3)

- Write the regression residuals as $T_i^{\perp \boldsymbol{X}} = \tilde{T}_i^*$ and $U_i^{\perp \boldsymbol{X}} = \tilde{U}_i$. Therefore,

$$\beta^* - \beta = \delta \times \frac{\mathrm{Cov}(U_i^{\perp \boldsymbol{X}}, T_i^{\perp \boldsymbol{X}})}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}.$$

- Applying FWL to the coefficient of U_i in the full model gives

$$\delta = \frac{\mathrm{Cov}(Y_i^{\perp T, \boldsymbol{X}}, U_i^{\perp T, \boldsymbol{X}})}{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]}.$$

- Therefore, the bias magnitude is

$$|\beta^* - \beta| = \underbrace{\frac{|\mathrm{Cov}(Y_i^{\perp T, \boldsymbol{X}}, U_i^{\perp T, \boldsymbol{X}})|}{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]}}_{=|\delta|} \times \frac{|\mathrm{Cov}(U_i^{\perp \boldsymbol{X}}, T_i^{\perp \boldsymbol{X}})|}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}.$$

Review: Unconditional R^2

- Recall that R^2 for regression $Y \sim \boldsymbol{X}$ is given by

$$R^2_{Y \sim \boldsymbol{X}} = \frac{\mathbb{V}[\hat{Y}_i]}{\mathbb{V}[Y_i]} = 1 - \frac{\mathbb{V}[\hat{\epsilon}_i]}{\mathbb{V}[Y_i]} = 1 - \frac{\mathbb{V}[Y_i^{\perp \boldsymbol{X}}]}{\mathbb{V}[Y_i]}.$$

- Now, notice that since residual $\hat{\epsilon}_i$ is orthogonal to $\hat{Y}_i$, we get

$$\mathrm{Cov}(Y_i, \hat{Y}_i) = \mathrm{Cov}(\hat{Y}_i + \hat{\epsilon}_i, \hat{Y}_i) = \mathrm{Cov}(\hat{Y}_i, \hat{Y}_i) = \mathbb{V}[\hat{Y}_i].$$

- As a result, we can show the connection between unconditional R^2 and the correlation coefficient:

$$\mathrm{Cor}(Y_i, \hat{Y}_i) = \frac{\mathrm{Cov}(Y_i, \hat{Y}_i)}{\sqrt{\mathbb{V}[Y_i]}\sqrt{\mathbb{V}[\hat{Y}_i]}} = \sqrt{\frac{\mathbb{V}[\hat{Y}_i]}{\mathbb{V}[Y_i]}} = \sqrt{R^2_{Y \sim \boldsymbol{X}}}.$$

Partial R^2 approach (1)

- We can do the same thing for partial R^2.
- Partial R^2 of $Y \sim U \mid T, \boldsymbol{X}$ is the R^2 from regressing $Y_i^{\perp T, \boldsymbol{X}}$ on $U_i^{\perp T, \boldsymbol{X}}$:

$$Y_i^{\perp T, \boldsymbol{X}} = \underbrace{\frac{\mathrm{Cov}(Y_i^{\perp T, \boldsymbol{X}}, U_i^{\perp T, \boldsymbol{X}})}{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]}}_{=\delta} U_i^{\perp T, \boldsymbol{X}} + \underbrace{\epsilon_i}_{Y_i^{\perp T, \boldsymbol{X}, U}}.$$

  We first remove what $(T_i, \boldsymbol{X}_i)$ explains from both variables.

- Now, notice that the fitted value and residual are orthogonal:

$$\mathbb{V}[Y_i^{\perp T, \boldsymbol{X}}] = \mathbb{V}[\delta U_i^{\perp T, \boldsymbol{X}}] + \mathbb{V}[\epsilon_i].$$

- Therefore, in the same way as unconditional R^2,

$$R^2_{Y \sim U \mid T, \boldsymbol{X}} = \frac{\mathbb{V}[\delta U_i^{\perp T, \boldsymbol{X}}]}{\mathbb{V}[Y_i^{\perp T, \boldsymbol{X}}]} = 1 - \frac{\mathbb{V}[\epsilon_i]}{\mathbb{V}[Y_i^{\perp T, \boldsymbol{X}}]} = 1 - \frac{\mathbb{V}[Y_i^{\perp T, \boldsymbol{X}, U}]}{\mathbb{V}[Y_i^{\perp T, \boldsymbol{X}}]}.$$

Partial R^2 approach (2)

- Substituting the FWL expression for δ, we can show the connection between partial R^2 and the correlation coefficient:

$$\begin{aligned} R^2_{Y\sim U|T,\boldsymbol{X}} &= \frac{\delta^2 \mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]} \\ &= \frac{\mathrm{Cov}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})^2}{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]} \\ &= \mathrm{Cor}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})^2. \end{aligned}$$

- The same argument gives

$$R^2_{T\sim U|\boldsymbol{X}} = \mathrm{Cor}(T_i^{\perp \boldsymbol{X}}, U_i^{\perp \boldsymbol{X}})^2 = R^2_{U\sim T|\boldsymbol{X}},$$

because correlation is symmetric. Thus the absolute partial correlation is the square root of partial R^2.

Partial R^2 approach (3)

- Now, notice that

$$|\delta| = \frac{|\mathrm{Cov}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})|}{\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]}$$

$$= \underbrace{\frac{|\mathrm{Cov}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})|}{\sqrt{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}\sqrt{\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]}}}_{=\sqrt{R^2_{Y\sim U|T,\boldsymbol{X}}}} \times \sqrt{\frac{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]}}.$$

- In the same way,

$$\frac{|\mathrm{Cov}(U_i^{\perp \boldsymbol{X}}, T_i^{\perp \boldsymbol{X}})|}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]} = \underbrace{\frac{|\mathrm{Cov}(U_i^{\perp \boldsymbol{X}}, T_i^{\perp \boldsymbol{X}})|}{\sqrt{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}\sqrt{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}}}_{=\sqrt{R^2_{T\sim U|\boldsymbol{X}}}} \times \sqrt{\frac{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}}.$$

Partial R^2 approach (4)

- Apply the same definition to $U \sim T \mid \boldsymbol{X}$:

$$R^2_{U \sim T \mid \boldsymbol{X}} = 1 - \frac{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}.$$

  The denominator is the variance remaining after removing $\boldsymbol{X}_i$. The numerator is the variance remaining after also removing T_i.

- Since $R^2_{U \sim T \mid \boldsymbol{X}} = R^2_{T \sim U \mid \boldsymbol{X}}$,

$$\frac{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]} = 1 - R^2_{U \sim T \mid \boldsymbol{X}} = 1 - R^2_{T \sim U \mid \boldsymbol{X}}.$$

- Taking reciprocals, for $R^2_{T \sim U \mid \boldsymbol{X}} < 1$,

$$\frac{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp T, \boldsymbol{X}}]} = \frac{1}{1 - R^2_{T \sim U \mid \boldsymbol{X}}}.$$

Partial R^2 approach (5)

- As a result, squaring the bias formula gives

$$
\begin{aligned}
|\beta^* - \beta|^2 &= R^2_{Y\sim U|T,\boldsymbol{X}} \frac{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]} \times R^2_{T\sim U|\boldsymbol{X}} \frac{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]} \\
&= R^2_{Y\sim U|T,\boldsymbol{X}} R^2_{T\sim U|\boldsymbol{X}} \frac{\mathbb{V}[U_i^{\perp \boldsymbol{X}}]}{\mathbb{V}[U_i^{\perp T,\boldsymbol{X}}]} \frac{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]} \\
&= \frac{R^2_{Y\sim U|T,\boldsymbol{X}} R^2_{T\sim U|\boldsymbol{X}}}{1 - R^2_{T\sim U|\boldsymbol{X}}} \frac{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}.
\end{aligned}
$$

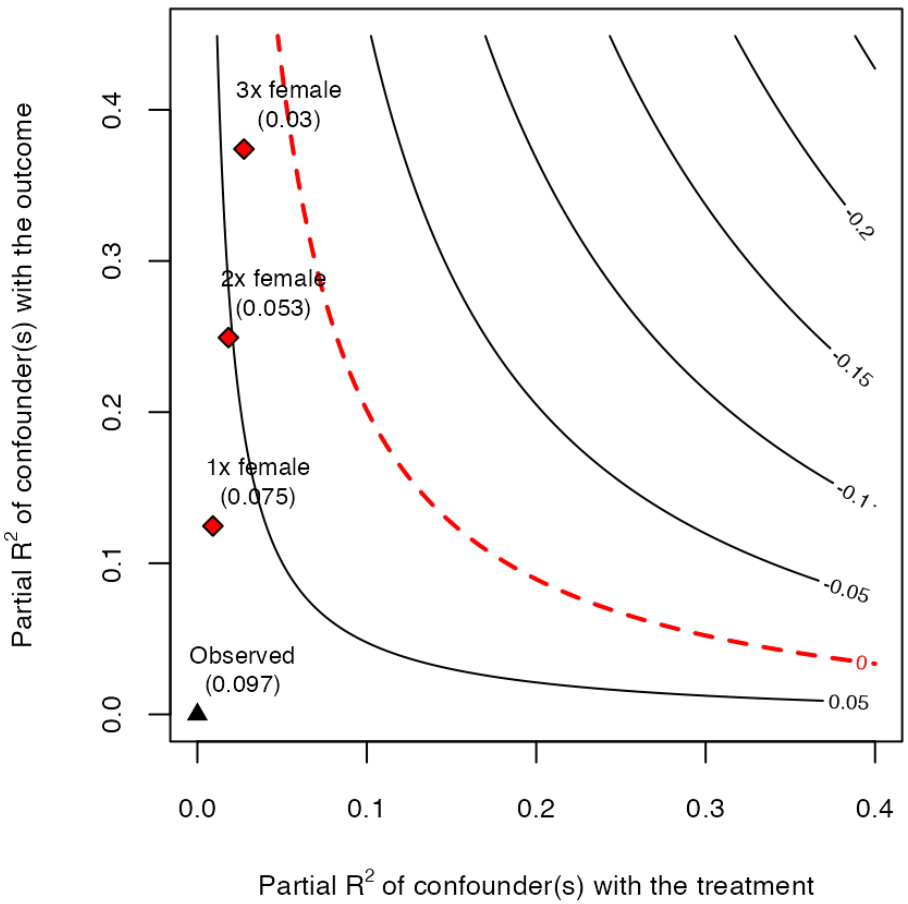
- Therefore, taking square roots, we get

$$
|\beta^* - \beta| = \sqrt{\frac{R^2_{Y\sim U|T,\boldsymbol{X}} R^2_{T\sim U|\boldsymbol{X}}}{1 - R^2_{T\sim U|\boldsymbol{X}}}} \times \underbrace{\sqrt{\frac{\mathbb{V}[Y_i^{\perp T,\boldsymbol{X}}]}{\mathbb{V}[T_i^{\perp \boldsymbol{X}}]}}}_{\text{Estimable}}.
$$

The bias direction depends on the signs of the associations.

Partial R^2 approach (6)

- Since we cannot directly observe the unobserved confounder, we typically benchmark the partial R^2 values using observed covariates.



Sensitivity Analysis of the Variance (1)

- Adding U_i also changes the estimated variance of the coefficient. Let $\widehat{\beta}^*$ and $\widehat{\beta}$ be the OLS coefficients without and with U_i, respectively.
- Here, residuals and partial R^2 values are sample quantities. The conventional (homoskedastic) OLS variance estimate is

$$\widehat{\mathbb{V}}(\widehat{\beta}^*) = \frac{\sum_{i=1}^n (Y_i^{\perp T, \boldsymbol{X}})^2 / \mathrm{df}}{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X}})^2}.$$

  The numerator estimates the error variance. By FWL, the denominator is the treatment variation remaining after controlling for $\boldsymbol{X}_i$.

- df is the residual degrees of freedom without U_i. Adding one scalar U_i leaves $\mathrm{df} - 1$ degrees of freedom:

$$\widehat{\mathbb{V}}(\widehat{\beta}) = \frac{\sum_{i=1}^n (Y_i^{\perp T, \boldsymbol{X}, U})^2 / (\mathrm{df} - 1)}{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X}, U})^2}.$$

Sensitivity Analysis of the Variance (2)

- Recall that partial R^2 measures the fraction of residual variation explained by U_i. Rearranging its definition gives

$$R^2_{Y\sim U|T,\boldsymbol{X}} = 1 - \frac{\sum_{i=1}^n (Y_i^{\perp T,\boldsymbol{X},U})^2}{\sum_{i=1}^n (Y_i^{\perp T,\boldsymbol{X}})^2},$$

$$\frac{\sum_{i=1}^n (Y_i^{\perp T,\boldsymbol{X},U})^2}{\sum_{i=1}^n (Y_i^{\perp T,\boldsymbol{X}})^2} = 1 - R^2_{Y\sim U|T,\boldsymbol{X}}.$$

- Applying the same definition to treatment gives

$$\frac{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X},U})^2}{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X}})^2} = 1 - R^2_{T\sim U|\boldsymbol{X}}.$$

- Taking reciprocals, for $R^2_{T\sim U|\boldsymbol{X}} < 1$,

$$\frac{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X}})^2}{\sum_{i=1}^n (T_i^{\perp \boldsymbol{X},U})^2} = \frac{1}{1 - R^2_{T\sim U|\boldsymbol{X}}}.$$

Sensitivity Analysis of the Variance (3)

- Dividing the two variance estimates, we obtain

$$\begin{aligned}\frac{\widehat{\mathbb{V}}(\widehat{\beta})}{\widehat{\mathbb{V}}(\widehat{\beta}^*)} &= \frac{\mathrm{df}}{\mathrm{df}-1}\frac{\sum_i (Y_i^{\perp T,\boldsymbol{X},U})^2}{\sum_i (Y_i^{\perp T,\boldsymbol{X}})^2}\frac{\sum_i (T_i^{\perp \boldsymbol{X}})^2}{\sum_i (T_i^{\perp \boldsymbol{X},U})^2} \\ &= \frac{\mathrm{df}}{\mathrm{df}-1}(1-R^2_{Y\sim U|T,\boldsymbol{X}})\frac{\sum_i (T_i^{\perp \boldsymbol{X}})^2}{\sum_i (T_i^{\perp \boldsymbol{X},U})^2} \\ &= \frac{\mathrm{df}}{\mathrm{df}-1}\frac{1-R^2_{Y\sim U|T,\boldsymbol{X}}}{1-R^2_{T\sim U|\boldsymbol{X}}}.\end{aligned}$$

- Taking square roots gives the adjusted standard error:

$$\widehat{\mathrm{se}}(\widehat{\beta}) = \widehat{\mathrm{se}}(\widehat{\beta}^*)\sqrt{\frac{\mathrm{df}}{\mathrm{df}-1}\frac{1-R^2_{Y\sim U|T,\boldsymbol{X}}}{1-R^2_{T\sim U|\boldsymbol{X}}}}.$$

- Other things equal, explaining more outcome variation lowers the standard error. Explaining more treatment variation raises it.

Cornfield Condition (Risk Ratio based approach)

- **Setup**: $Y_i(t) \perp\!\!\!\perp T_i \mid U_i$ for $t \in \{0, 1\}$
  - However, U_i is unobserved
- **Estimand**: Now, let's focus on the **causal risk ratio**:

$$RR_{TY}^{\text{true}} = \frac{\mathbb{P}(Y_i(1) = 1)}{\mathbb{P}(Y_i(0) = 1)}$$

  - Risk ratio $= 1$ is equivalent to ATE $= 0$
- We instead observe the observed risk ratio

$$RR_{TY}^{\text{obs}} = \frac{\mathbb{P}(Y_i = 1 \mid T_i = 1)}{\mathbb{P}(Y_i = 1 \mid T_i = 0)}$$

Cornfield Condition (Risk Ratio based approach)

- **Generalized Cornfield Condition**: If $\mathrm{RR}_{TY}^{\mathrm{obs}} > 1$, then

$$RR_{TY}^{\mathrm{true}} \geq \mathrm{RR}_{TY}^{\mathrm{obs}} \times \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU} \times \mathrm{RR}_{UY}}$$

where

$$\mathrm{RR}_{TU} = \frac{\mathbb{P}(U_i = 1 \mid T_i = 1)}{\mathbb{P}(U_i = 1 \mid T_i = 0)}, \quad \mathrm{RR}_{UY} = \frac{\mathbb{P}(Y_i = 1 \mid U_i = 1)}{\mathbb{P}(Y_i = 1 \mid U_i = 0)}$$

Further, in order for $RR_{TY}^{\mathrm{true}} = 1$, we must have

$$\underbrace{\max\{\mathrm{RR}_{UY}, \mathrm{RR}_{TU}\}}_{\text{Unobserved}} \geq \underbrace{\mathrm{RR}_{TY}^{\mathrm{obs}} + \sqrt{\mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TY}^{\mathrm{obs}} - 1)}}_{\text{Observed}}$$

  - This is tighter than **cornfield condition**, which states that

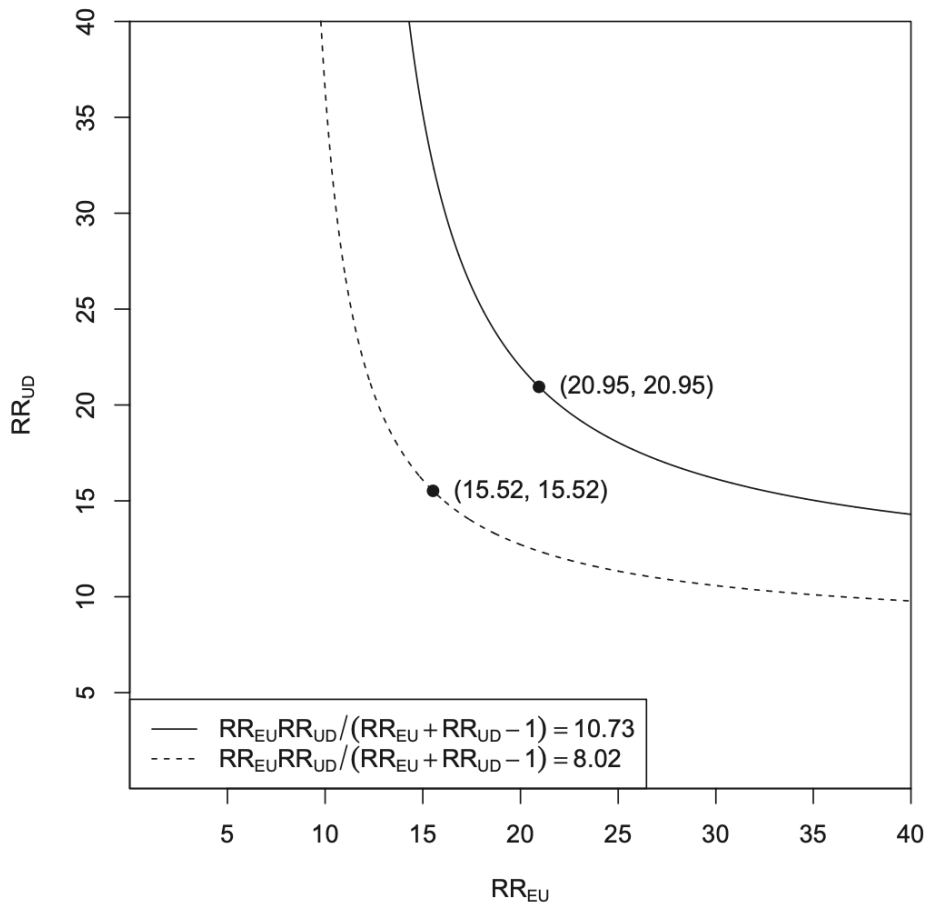
$$\min\{\mathrm{RR}_{UY}, \mathrm{RR}_{TU}\} \geq \mathrm{RR}_{TY}^{\mathrm{obs}}$$

  - Note that cornfield condition is about minimum, while generalized cornfield condition is about maximum.

Cornfield Condition (Risk Ratio based approach)

- Recall that if $\mathrm{RR}_{TY}^{\mathrm{obs}} > 1$, then

$$RR_{TY}^{\mathrm{true}} \times \frac{\mathrm{RR}_{TU} \times \mathrm{RR}_{UY}}{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1} \geq \mathrm{RR}_{TY}^{\mathrm{obs}}$$



Derivation of Cornfield Condition (1)

- Suppose treatment has no effect within each stratum of U_i:

  $$\Pr\{Y_i(1) = 1 \mid U_i = u\} = \Pr\{Y_i(0) = 1 \mid U_i = u\} =: r_u, \qquad u \in \{0, 1\}.$$

  Averaging over U_i gives $\mathrm{RR}_{TY}^{\mathrm{true}} = 1$.

- Then, for each $t \in \{0, 1\}$,

  $$\begin{aligned}\Pr(Y_i = 1 \mid T_i = t, U_i = u) &= \Pr\{Y_i(t) = 1 \mid T_i = t, U_i = u\} \\ &= \Pr\{Y_i(t) = 1 \mid U_i = u\} = r_u.\end{aligned}$$

- This does not depend on t, so $T_i \perp\!\!\!\perp Y_i \mid U_i$
  - This makes sense since treatment has no effect. Once you fix U, T and Y are uncorrelated.

- Averaging over T_i within $U_i = u$ then gives $\Pr(Y_i = 1 \mid U_i = u) = r_u$.

Derivation of Cornfield Condition (2)

- Let $p_t := \Pr(U_i = 1 \mid T_i = t)$. With the definitions from before,

$$\mathrm{RR}_{TU} = \frac{p_1}{p_0}, \qquad \mathrm{RR}_{UY} = \frac{\Pr(Y_i = 1 \mid U_i = 1)}{\Pr(Y_i = 1 \mid U_i = 0)} = \frac{r_1}{r_0}.$$

  Code U_i so that $\mathrm{RR}_{UY} \geq 1$ (else relabel it as $1 - U_i$); assume $p_0, r_0 > 0$.

- By total probability,

$$\Pr(Y_i = 1 \mid T_i = t) = r_1 p_t + r_0(1 - p_t) = r_0\{1 + (\mathrm{RR}_{UY} - 1)p_t\}.$$

- Dividing the treated risk by the control risk, r_0 cancels:

$$\mathrm{RR}_{TY}^{\mathrm{obs}} = \frac{1 + (\mathrm{RR}_{UY} - 1)p_1}{1 + (\mathrm{RR}_{UY} - 1)p_0}.$$

- Subtracting 1,

$$\mathrm{RR}_{TY}^{\mathrm{obs}} - 1 = \frac{(\mathrm{RR}_{UY} - 1)(p_1 - p_0)}{1 + (\mathrm{RR}_{UY} - 1)p_0}.$$

  The denominator is at least 1. So $\mathrm{RR}_{TY}^{\mathrm{obs}} > 1$ requires $\mathrm{RR}_{UY} > 1$ and $p_1 > p_0$, i.e., $\mathrm{RR}_{TU} > 1$.

Derivation of Cornfield Condition (3)

- **Treatment–confounder association.** Suppose $\mathrm{RR}_{TY}^{\mathrm{obs}} > 1$. Then

$$\begin{aligned}
\mathrm{RR}_{TU} - \mathrm{RR}_{TY}^{\mathrm{obs}} &= \frac{p_1}{p_0} - \frac{1 + (\mathrm{RR}_{UY} - 1)p_1}{1 + (\mathrm{RR}_{UY} - 1)p_0} \\
&= \frac{p_1\{1 + (\mathrm{RR}_{UY} - 1)p_0\} - p_0\{1 + (\mathrm{RR}_{UY} - 1)p_1\}}{p_0\{1 + (\mathrm{RR}_{UY} - 1)p_0\}} \\
&= \frac{p_1 - p_0}{p_0\{1 + (\mathrm{RR}_{UY} - 1)p_0\}} > 0.
\end{aligned}$$

  The last step uses $p_1 > p_0$ from the previous slide.

- Therefore, $\mathrm{RR}_{TU} \geq \mathrm{RR}_{TY}^{\mathrm{obs}}$.

- Intuition: $\Pr(Y_i = 1 \mid T_i = t) = r_0 + (r_1 - r_0)p_t$. Adding the same baseline risk r_0 to the treated and control risks pulls the ratio p_1/p_0 toward 1.

Derivation of Cornfield Condition (4)

- **Confounder–outcome association.** Since $\mathrm{RR}_{UY} - 1 \geq 0$, $p_1 \leq 1$, and $p_0 \geq 0$,

$$\mathrm{RR}_{TY}^{\mathrm{obs}} = \frac{1 + (\mathrm{RR}_{UY} - 1)p_1}{1 + (\mathrm{RR}_{UY} - 1)p_0} \leq \frac{1 + (\mathrm{RR}_{UY} - 1) \times 1}{1 + (\mathrm{RR}_{UY} - 1) \times 0} = \mathrm{RR}_{UY}.$$

- Combining the two bounds gives the **Cornfield condition**:

$$\min\{\mathrm{RR}_{TU}, \mathrm{RR}_{UY}\} \geq \mathrm{RR}_{TY}^{\mathrm{obs}}.$$

  To explain away $\mathrm{RR}_{TY}^{\mathrm{obs}}$, U_i must be at least as strongly associated with *both* T_i and Y_i as T_i is with Y_i.

- This bounds each association *separately*. The generalized Cornfield condition bounds them *jointly*, which makes it tighter.

Extra: Derivation of Generalized Cornfield Condition (1)

- By $Y_i(t) \perp\!\!\!\perp T_i \mid U_i$ and consistency,

$$\begin{aligned}\Pr\{Y_i(t) = 1 \mid U_i = u\} &= \Pr\{Y_i(t) = 1 \mid T_i = t, U_i = u\} \\ &= \Pr(Y_i = 1 \mid T_i = t, U_i = u).\end{aligned}$$

  Assume common support and positive outcome risks.

- By total probability, for $s \in \{0, 1\}$,

$$\Pr\{Y_i(t) = 1 \mid T_i = s\} = \sum_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \Pr(U_i = u \mid T_i = s).$$

  Also,

$$\Pr\{Y_i(t) = 1\} = \sum_{s=0}^{1} \Pr\{Y_i(t) = 1 \mid T_i = s\} \Pr(T_i = s).$$

- For the treatment each group actually received,

$$\Pr\{Y_i(t) = 1 \mid T_i = t\} = \Pr(Y_i = 1 \mid T_i = t).$$

Extra: Derivation of Generalized Cornfield Condition (2)

- Define the two sensitivity parameters:

$$\mathrm{RR}_{TU} := \max_u \frac{\Pr(U_i = u \mid T_i = 1)}{\Pr(U_i = u \mid T_i = 0)},$$

$$\mathrm{RR}_{UY} := \max_{t \in \{0,1\}} \frac{\max_u \Pr\{Y_i(t) = 1 \mid U_i = u\}}{\min_u \Pr\{Y_i(t) = 1 \mid U_i = u\}}.$$

- Both risk ratios are at least 1.
  - For RR_{TU},

$$1 = \sum_u \Pr(U_i = u \mid T_i = 1) \leq \mathrm{RR}_{TU} \sum_u \Pr(U_i = u \mid T_i = 0) = \mathrm{RR}_{TU}.$$

    Both probability distributions sum to 1.
  - For RR_{UY}, the largest outcome risk is at least the smallest, so for either t,

$$\frac{\max_u \Pr\{Y_i(t) = 1 \mid U_i = u\}}{\min_u \Pr\{Y_i(t) = 1 \mid U_i = u\}} \geq 1.$$

    Taking the maximum over t gives $\mathrm{RR}_{UY} \geq 1$.

Extra: Derivation of Generalized Cornfield Condition (3)

- For either treatment value t,

$$\begin{aligned}\Pr\{Y_i(t) = 1 \mid T_i = 1\} &= \sum_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \Pr(U_i = u \mid T_i = 1) \\ &\leq \max_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \\ &\leq \mathrm{RR}_{UY} \min_u \Pr\{Y_i(t) = 1 \mid U_i = u\}.\end{aligned}$$

  where the first inequality bounds a weighted average by its largest value.

Extra: Derivation of Generalized Cornfield Condition (4)

- By definition of RR_{TU}, for every u,

$$\mathrm{RR}_{TU} \Pr(U_i = u \mid T_i = 0) - \Pr(U_i = u \mid T_i = 1) \geq 0.$$

- Therefore,

$$\begin{aligned}
&\mathrm{RR}_{TU} \Pr\{Y_i(t) = 1 \mid T_i = 0\} - \Pr\{Y_i(t) = 1 \mid T_i = 1\} \\
&\quad = \sum_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \\
&\qquad \times \underbrace{\{\mathrm{RR}_{TU} \Pr(U_i = u \mid T_i = 0) - \Pr(U_i = u \mid T_i = 1)\}}_{\geq 0} \\
&\quad \geq \min_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \\
&\qquad \times \{\mathrm{RR}_{TU} \sum_u \Pr(U_i = u \mid T_i = 0) - \sum_u \Pr(U_i = u \mid T_i = 1)\} \\
&\quad = (\mathrm{RR}_{TU} - 1) \min_u \Pr\{Y_i(t) = 1 \mid U_i = u\}.
\end{aligned}$$

The last equality uses $\sum_u \Pr(U_i = u \mid T_i = s) = 1$ for either $s \in \{0, 1\}$.

Extra: Derivation of Generalized Cornfield Condition (5)

- Recall the outcome-risk bound derived earlier:

$$\min_u \Pr\{Y_i(t) = 1 \mid U_i = u\} \geq \frac{\Pr\{Y_i(t) = 1 \mid T_i = 1\}}{\mathrm{RR}_{UY}}.$$

- Since $\mathrm{RR}_{TU} - 1 \geq 0$, substituting this into the preceding slide gives

$$\begin{aligned}
&\mathrm{RR}_{TU} \Pr\{Y_i(t) = 1 \mid T_i = 0\} - \Pr\{Y_i(t) = 1 \mid T_i = 1\} \\
&\quad \geq \frac{\mathrm{RR}_{TU} - 1}{\mathrm{RR}_{UY}} \Pr\{Y_i(t) = 1 \mid T_i = 1\}.
\end{aligned}$$

  Adding $\Pr\{Y_i(t) = 1 \mid T_i = 1\}$ to both sides,

$$\begin{aligned}
&\mathrm{RR}_{TU} \Pr\{Y_i(t) = 1 \mid T_i = 0\} \\
&\quad \geq \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{UY}} \Pr\{Y_i(t) = 1 \mid T_i = 1\}.
\end{aligned}$$

- Dividing by the positive terms and rearranging gives

$$\frac{\Pr\{Y_i(t) = 1 \mid T_i = 1\}}{\Pr\{Y_i(t) = 1 \mid T_i = 0\}} \leq \frac{\mathrm{RR}_{TU}\mathrm{RR}_{UY}}{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}.$$

Extra: Derivation of Generalized Cornfield Condition (6)

- By consistency and the definition of $\mathrm{RR}_{TY}^{\mathrm{obs}}$,

$$\Pr\{Y_i(1) = 1 \mid T_i = 1\} = \mathrm{RR}_{TY}^{\mathrm{obs}} \Pr\{Y_i(0) = 1 \mid T_i = 0\}.$$

- For controls ($T_i = 0$), set $t = 1$ in the previous bound and rearrange:

$$\begin{aligned}
&\Pr\{Y_i(1) = 1 \mid T_i = 0\} \\
&\quad \geq \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}} \Pr\{Y_i(1) = 1 \mid T_i = 1\} \\
&\quad = \mathrm{RR}_{TY}^{\mathrm{obs}} \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}} \Pr\{Y_i(0) = 1 \mid T_i = 0\}.
\end{aligned}$$

- For treated units ($T_i = 1$), use the same bound with $t = 0$:

$$\begin{aligned}
&\Pr\{Y_i(1) = 1 \mid T_i = 1\} \\
&\quad = \mathrm{RR}_{TY}^{\mathrm{obs}} \Pr\{Y_i(0) = 1 \mid T_i = 0\} \\
&\quad \geq \mathrm{RR}_{TY}^{\mathrm{obs}} \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}} \Pr\{Y_i(0) = 1 \mid T_i = 1\}.
\end{aligned}$$

Extra: Derivation of Generalized Cornfield Condition (7)

- Therefore, for each $s \in \{0, 1\}$,

$$\Pr\{Y_i(1) = 1 \mid T_i = s\} \geq \mathrm{RR}_{TY}^{\mathrm{obs}} \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}} \Pr\{Y_i(0) = 1 \mid T_i = s\}.$$

- Multiply by $\Pr(T_i = s)$ and sum over s. By total probability,

$$\Pr\{Y_i(1) = 1\} \geq \mathrm{RR}_{TY}^{\mathrm{obs}} \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}} \Pr\{Y_i(0) = 1\}.$$

- Dividing by $\Pr\{Y_i(0) = 1\}$ gives

$$\mathrm{RR}_{TY}^{\mathrm{true}} \geq \mathrm{RR}_{TY}^{\mathrm{obs}} \frac{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}{\mathrm{RR}_{TU}\mathrm{RR}_{UY}}.$$

- This is the first part of the generalized Cornfield condition.

Extra: Derivation of Generalized Cornfield Condition (8)

- Suppose that $\mathrm{RR}_{TY}^{\mathrm{obs}} > 1$ but $\mathrm{RR}_{TY}^{\mathrm{true}} = 1$. Then,

$$\mathrm{RR}_{TY}^{\mathrm{obs}} \leq \frac{\mathrm{RR}_{TU}\mathrm{RR}_{UY}}{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1}.$$

  - This is considering the case where you find the association but the true causal effect is null.

- As sensitivity parameters are at least 1, $\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1 \geq \max\{\mathrm{RR}_{TU}, \mathrm{RR}_{UY}\}$. So,

$$\mathrm{RR}_{TY}^{\mathrm{obs}} \leq \frac{\mathrm{RR}_{TU}\mathrm{RR}_{UY}}{\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1} \leq \min\{\mathrm{RR}_{TU}, \mathrm{RR}_{UY}\}.$$

- Multiply by $\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1$ and rearrange:

$$\mathrm{RR}_{TU}\mathrm{RR}_{UY} \geq \mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TU} + \mathrm{RR}_{UY} - 1),$$

$$(\mathrm{RR}_{TU} - \mathrm{RR}_{TY}^{\mathrm{obs}})(\mathrm{RR}_{UY} - \mathrm{RR}_{TY}^{\mathrm{obs}}) \geq \mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TY}^{\mathrm{obs}} - 1).$$

  The second line subtracts $\mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TU} + \mathrm{RR}_{UY})$ and adds $(\mathrm{RR}_{TY}^{\mathrm{obs}})^2$ to both sides.

Extra: Derivation of Generalized Cornfield Condition (9)

- Both differences in the previous product are nonnegative. Replacing each association by the larger one can only increase it:

$$\begin{aligned}\left[\max\{\mathrm{RR}_{TU}, \mathrm{RR}_{UY}\} - \mathrm{RR}_{TY}^{\mathrm{obs}}\right]^2 &\geq (\mathrm{RR}_{TU} - \mathrm{RR}_{TY}^{\mathrm{obs}})(\mathrm{RR}_{UY} - \mathrm{RR}_{TY}^{\mathrm{obs}}) \\ &\geq \mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TY}^{\mathrm{obs}} - 1).\end{aligned}$$

- Taking square roots and adding $\mathrm{RR}_{TY}^{\mathrm{obs}}$ gives

$$\max\{\mathrm{RR}_{TU}, \mathrm{RR}_{UY}\} \geq \mathrm{RR}_{TY}^{\mathrm{obs}} + \sqrt{\mathrm{RR}_{TY}^{\mathrm{obs}}(\mathrm{RR}_{TY}^{\mathrm{obs}} - 1)}.$$

- This is the second part of the generalized Cornfield condition.

Partial Identification

- **Partial Identification**: How much can we know with the minimal amount of assumptions we are willing to make?
  - The more we make assumptions, the more we gains
  - The credibility of inference decreases with the strength of the assumptions maintained

- **Tips**
  - Write down all the constraints (sum of probability, probability is non-negative, etc)
  - Combine every constraints to create the bound

No Assumption Bounds for the ATE (1)

- Consider binary treatment and outcome, with $0 < \mathbb{P}(T_i = 1) < 1$. Our quantity of interest is

$$\tau = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

  We maintain consistency, $Y_i = Y_i(T_i)$, but do not assume unconfoundedness.

- By iterated expectations and consistency,

$$\begin{aligned}\mathbb{E}[Y_i(1)] &= \mathbb{E}[Y_i(1) \mid T_i = 1]\mathbb{P}(T_i = 1) + \mathbb{E}[Y_i(1) \mid T_i = 0]\mathbb{P}(T_i = 0) \\ &= \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) + \underbrace{\mathbb{E}[Y_i(1) \mid T_i = 0]}_{\text{Unobserved}}\mathbb{P}(T_i = 0).\end{aligned}$$

- Since $Y_i(1)$ is binary, $0 \leq \mathbb{E}[Y_i(1) \mid T_i = 0] \leq 1$.

No Assumption Bounds for the ATE (2)

- Replacing the missing mean by 0 or 1 gives

$$\mathbb{E}[Y_i(1)] \geq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1),$$
$$\mathbb{E}[Y_i(1)] \leq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) + \mathbb{P}(T_i = 0).$$

- For the control outcome, the same decomposition gives

$$\mathbb{E}[Y_i(0)] = \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) + \underbrace{\mathbb{E}[Y_i(0) \mid T_i = 1]}_{\in[0,1]}\mathbb{P}(T_i = 1).$$

- Therefore,

$$\mathbb{E}[Y_i(0)] \geq \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0),$$
$$\mathbb{E}[Y_i(0)] \leq \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) + \mathbb{P}(T_i = 1).$$

No Assumption Bounds for the ATE (3)

- To minimize τ, subtract the largest possible $\mathbb{E}[Y_i(0)]$ from the smallest possible $\mathbb{E}[Y_i(1)]$:

$$\begin{aligned}\tau &\geq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) - \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) - \mathbb{P}(T_i = 1) \\ &= -\mathbb{P}(Y_i = 0 \mid T_i = 1)\mathbb{P}(T_i = 1) - \mathbb{P}(Y_i = 1 \mid T_i = 0)\mathbb{P}(T_i = 0) \\ &= -\mathbb{P}(Y_i = 0, T_i = 1) - \mathbb{P}(Y_i = 1, T_i = 0)\end{aligned}$$

- To maximize τ, reverse these choices:

$$\begin{aligned}\tau &\leq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) + \mathbb{P}(T_i = 0) - \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) \\ &= \mathbb{P}(Y_i = 1 \mid T_i = 1)\mathbb{P}(T_i = 1) + \mathbb{P}(Y_i = 0 \mid T_i = 0)\mathbb{P}(T_i = 0) \\ &= \mathbb{P}(Y_i = 1, T_i = 1) + \mathbb{P}(Y_i = 0, T_i = 0)\end{aligned}$$

ATE Bounds with Monotonicity: Perfect Selection (1)

- Continue with binary treatment and outcome. Suppose

$$\mathbb{E}[Y_i(1) - Y_i(0) \mid T_i = 1] \geq 0,$$
$$\mathbb{E}[Y_i(1) - Y_i(0) \mid T_i = 0] \leq 0.$$

  This is perfect treatment selection: on average, treated units do at least as well with treatment, while untreated units do at least as well without it.
- By consistency, these inequalities imply

$$\mathbb{E}[Y_i(0) \mid T_i = 1] \leq \mathbb{E}[Y_i(1) \mid T_i = 1] = \mathbb{E}[Y_i \mid T_i = 1],$$
$$\mathbb{E}[Y_i(1) \mid T_i = 0] \leq \mathbb{E}[Y_i(0) \mid T_i = 0] = \mathbb{E}[Y_i \mid T_i = 0].$$

- Since the potential outcomes are binary, the missing means satisfy

$$0 \leq \mathbb{E}[Y_i(0) \mid T_i = 1] \leq \mathbb{E}[Y_i \mid T_i = 1],$$
$$0 \leq \mathbb{E}[Y_i(1) \mid T_i = 0] \leq \mathbb{E}[Y_i \mid T_i = 0].$$

ATE Bounds with Monotonicity: Perfect Selection (2)

- By iterated expectations and consistency,

$$\begin{aligned}\tau = &\big\{\mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i(0) \mid T_i = 1]\big\}\mathbb{P}(T_i = 1) \\ &+ \big\{\mathbb{E}[Y_i(1) \mid T_i = 0] - \mathbb{E}[Y_i \mid T_i = 0]\big\}\mathbb{P}(T_i = 0).\end{aligned}$$

- To minimize τ, set the missing control mean among treated units to its upper bound and the missing treatment mean among untreated units to 0:

$$\begin{aligned}\tau &\geq 0 \times \mathbb{P}(T_i = 1) - \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) \\ &= -\mathbb{P}(Y_i = 1, T_i = 0).\end{aligned}$$

- Reversing these choices gives the upper bound:

$$\begin{aligned}\tau &\leq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) + 0 \times \mathbb{P}(T_i = 0) \\ &= \mathbb{P}(Y_i = 1, T_i = 1).\end{aligned}$$

These bounds are tighter than the no-assumption bounds, but still include 0.

ATE Bounds with Monotonicity: Monotone Selection (1)

- A different assumption is monotone treatment selection:

$$\mathbb{E}[Y_i(t) \mid T_i = 1] \geq \mathbb{E}[Y_i(t) \mid T_i = 0], \qquad t \in \{0, 1\}.$$

  For example, those who obtain more schooling may have higher wage potential under either schooling level.

- Return to binary outcomes. For $t = 1$, consistency gives

$$0 \leq \mathbb{E}[Y_i(1) \mid T_i = 0] \leq \mathbb{E}[Y_i(1) \mid T_i = 1] = \mathbb{E}[Y_i \mid T_i = 1].$$

  For $t = 0$, it gives

$$\mathbb{E}[Y_i \mid T_i = 0] = \mathbb{E}[Y_i(0) \mid T_i = 0] \leq \mathbb{E}[Y_i(0) \mid T_i = 1] \leq 1.$$

- This orders the two treatment groups under the same treatment. It does not require treatment to help every individual.

ATE Bounds with Monotonicity: Monotone Selection (2)

- Substituting the upper bound for the missing treatment mean,

$$\begin{aligned}\mathbb{E}[Y_i(1)] &\leq \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 1) + \mathbb{E}[Y_i \mid T_i = 1]\mathbb{P}(T_i = 0) \\ &= \mathbb{E}[Y_i \mid T_i = 1]. \\ \mathbb{E}[Y_i(0)] &\geq \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) + \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 1) \\ &= \mathbb{E}[Y_i \mid T_i = 0].\end{aligned}$$

- Therefore, the observed difference is an upper bound for the ATE:

$$\tau \leq \mathbb{E}[Y_i \mid T_i = 1] - \mathbb{E}[Y_i \mid T_i = 0].$$

- The no-assumption lower bound is still attainable.
  - Set $Y_i(1) = 0$ for untreated units and $Y_i(0) = 1$ for treated units.
  - Substituting these values gives

$$\begin{aligned}\tau &\geq \{\mathbb{E}[Y_i \mid T_i = 1] - 1\}\mathbb{P}(T_i = 1) - \mathbb{E}[Y_i \mid T_i = 0]\mathbb{P}(T_i = 0) \\ &= -\mathbb{P}(Y_i = 0, T_i = 1) - \mathbb{P}(Y_i = 1, T_i = 0).\end{aligned}$$

  - Thus monotone selection tightens the upper bound, while leaving the lower bound unchanged.

Partial Identification: Case of Binary Outcome (1)

- Let's think about the binary outcome and treatment. We have the following principal strata:

$$(Y_i(0), Y_i(1)) \in \{(0,0), (0,1), (1,0), (1,1)\}$$

- Suppose that we want to assign treatment to maximize the effect $Y_i(1) - Y_i(0)$
  - That is, assigning treatment to the strata $(Y_i(0), Y_i(1)) = (0,1)$ and not assigning to the strata $(Y_i(0), Y_i(1)) = (1,0)$
  - The only people whose outcome is 0 is those in strata $(Y_i(0), Y_i(1)) = (0,0)$

- **Question**: How can we maximize the outcome value by optimizing the treatment assignment?

- If we optimally assign the treatment effect, the observed outcome δ will be

$$\begin{aligned}\delta :=& 1 \times \mathbb{P}(Y_i(0) = 1, Y_i(1) = 1) + 1 \times \mathbb{P}(Y_i(0) = 0, Y_i(1) = 1) \\ &+ 1 \times \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0) + 0 \times \mathbb{P}(Y_i(0) = 0, Y_i(1) = 0)\end{aligned}$$

Partial Identification: Case of Binary Outcome (2)

- Now,

$$\mathbb{P}(Y_i(1) = 1) = \mathbb{P}(Y_i(0) = 0, Y_i(1) = 1) + \mathbb{P}(Y_i(0) = 1, Y_i(1) = 1)$$
$$\underbrace{\mathbb{P}(Y_i(0) = 1)}_{\text{Identifiable}} = \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0) + \mathbb{P}(Y_i(0) = 1, Y_i(1) = 1)$$

  but we do not observe the probability of each principal strata.

- But we know that

$$\delta = \underbrace{\mathbb{P}(Y_i(1) = 1)}_{\text{Identifiable}} + \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0)$$

  so we need to think about how to maximize $\mathbb{P}(Y_i(0) = 1, Y_i(1) = 0)$

Partial Identification: Case of Binary Outcome (3)

- Let's write down all the constraints:
  - Firstly, each probability is bounded between 0 and 1
  - Then, we can identify $\mathbb{P}(Y_i(1) = 1)$ and $\mathbb{P}(Y_i(0) = 1)$
- In this case, each strata probability can be written as observed quantity and $\mathbb{P}(Y_i(0) = 1, Y_i(1) = 0)$. I.e.,

$$0 \leq \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0) \leq 1$$
$$0 \leq \underbrace{\mathbb{P}(Y_i = 1 \mid T_i = 0) - \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0)}_{=\mathbb{P}(Y_i(0)=1, Y_i(1)=1)} \leq 1$$

- You can also derive $\mathbb{P}(Y_i(0) = 0, Y_i(1) = 1)$ and $\mathbb{P}(Y_i(0) = 0, Y_i(1) = 0)$
- Under these constraints, think about how much you can maximize the quantity of interest.

Partial Identification: IV and ATE (1)

- Consider binary instrument Z_i, treatment T_i, and outcome Y_i. For example, an offer of job training, participation, and employment.
- Let $T_i(z)$ denote treatment received under instrument value z. Let $Y_i(t, z)$ denote the outcome if we set treatment to t and the instrument to z.
- Under exclusion, $Y_i(t, z) = Y_i(t)$. Our quantity of interest is the population ATE:

$$\tau = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

- Treatment uptake may be confounded. Our goal is to use a randomized instrument to bound τ.

Partial Identification: IV and ATE (2)

- **Randomization**: the instrument is independent of all potential treatments and outcomes:

$$Z_i \perp\!\!\!\perp \big(T_i(0), T_i(1), \{Y_i(t,z) : t,z \in \{0,1\}\}\big).$$

  We also require $0 < \mathbb{P}(Z_i = 1) < 1$.

- **Exclusion restriction**: holding treatment fixed, changing the instrument does not change the outcome:

$$Y_i(t,1) = Y_i(t,0) =: Y_i(t), \qquad t \in \{0,1\}.$$

- **Consistency**: observed treatment and outcome equal their potential values under the realized assignments:

$$T_i = T_i(Z_i), \qquad Y_i = Y_i(T_i, Z_i).$$

  Under exclusion, $Y_i = Y_i(T_i)$.

- These bounds require neither treatment monotonicity nor instrument relevance (so we cannot use standard IV methods).

Partial Identification: IV and ATE (3)

- Among units with $T_i = 1$ and $Z_i = 1$,

$$Y_i = \underbrace{Y_i(1,1)}_{\text{Consistency}} = \underbrace{Y_i(1)}_{\text{Exclusion}} .$$

  Among units with $T_i = 1$ and $Z_i = 0$,

$$Y_i = \underbrace{Y_i(1,0)}_{\text{Consistency}} = \underbrace{Y_i(1)}_{\text{Exclusion}} .$$

- Exclusion lets both instrument groups reveal the same treatment potential outcome. Without it, $Y_i(1,1)$ and $Y_i(1,0)$ can differ.
- Once we write the common outcome as $Y_i(1)$, replacing it with Y_i among treated units uses consistency. Randomization then gives

$$\mathbb{P}(Y_i(1) = 1) = \mathbb{P}(Y_i(1) = 1 \mid Z_i = z), \qquad z \in \{0,1\}.$$

Partial Identification: IV and ATE (4)

- First, use the $Z_i = 1$ group to learn about $Y_i(1)$:

$$\begin{aligned}
\mathbb{P}(Y_i(1) = 1) &= \mathbb{P}(Y_i(1) = 1 \mid Z_i = 1) && \text{(Randomization)} \\
&= \mathbb{P}(Y_i(1) = 1, T_i = 1 \mid Z_i = 1) \\
&\quad + \mathbb{P}(Y_i(1) = 1, T_i = 0 \mid Z_i = 1) && \text{(Total probability)} \\
&= \mathbb{P}(Y_i = 1, T_i = 1 \mid Z_i = 1) \\
&\quad + \underbrace{\mathbb{P}(Y_i(1) = 1, T_i = 0 \mid Z_i = 1)}_{\text{Unobserved}} && \text{(Consistency).}
\end{aligned}$$

- The unobserved probability lies between 0 and $\mathbb{P}(T_i = 0 \mid Z_i = 1)$. Therefore,

$$\begin{aligned}
&\mathbb{P}(Y_i(1) = 1) \geq \mathbb{P}(Y_i = 1, T_i = 1 \mid Z_i = 1), \\
&\mathbb{P}(Y_i(1) = 1) \leq \mathbb{P}(Y_i = 1, T_i = 1 \mid Z_i = 1) + \mathbb{P}(T_i = 0 \mid Z_i = 1).
\end{aligned}$$

Partial Identification: IV and ATE (5)

- Next, use the $Z_i = 0$ group to learn about $Y_i(0)$:

$$\begin{aligned}
\mathbb{P}(Y_i(0) = 1) &= \mathbb{P}(Y_i(0) = 1 \mid Z_i = 0) && \text{(Randomization)} \\
&= \mathbb{P}(Y_i(0) = 1, T_i = 0 \mid Z_i = 0) \\
&\quad + \mathbb{P}(Y_i(0) = 1, T_i = 1 \mid Z_i = 0) && \text{(Total probability)} \\
&= \mathbb{P}(Y_i = 1, T_i = 0 \mid Z_i = 0) \\
&\quad + \underbrace{\mathbb{P}(Y_i(0) = 1, T_i = 1 \mid Z_i = 0)}_{\text{Unobserved}} && \text{(Consistency).}
\end{aligned}$$

- The unobserved probability lies between 0 and $\mathbb{P}(T_i = 1 \mid Z_i = 0)$. Therefore,

$$\begin{aligned}
&\mathbb{P}(Y_i(0) = 1) \geq \mathbb{P}(Y_i = 1, T_i = 0 \mid Z_i = 0), \\
&\mathbb{P}(Y_i(0) = 1) \leq \mathbb{P}(Y_i = 1, T_i = 0 \mid Z_i = 0) + \mathbb{P}(T_i = 1 \mid Z_i = 0).
\end{aligned}$$

Partial Identification: IV and ATE (6)

- To minimize τ, use the lower bound for $\mathbb{E}[Y_i(1)]$ and the upper bound for $\mathbb{E}[Y_i(0)]$:

$$\begin{aligned}\tau \geq\ & \mathbb{P}(Y_i = 1, T_i = 1 \mid Z_i = 1) \\ & - \mathbb{P}(Y_i = 1, T_i = 0 \mid Z_i = 0) - \mathbb{P}(T_i = 1 \mid Z_i = 0).\end{aligned}$$

- To maximize τ, use the upper bound for $\mathbb{E}[Y_i(1)]$ and the lower bound for $\mathbb{E}[Y_i(0)]$:

$$\begin{aligned}\tau \leq\ & \mathbb{P}(Y_i = 1, T_i = 1 \mid Z_i = 1) + \mathbb{P}(T_i = 0 \mid Z_i = 1) \\ & - \mathbb{P}(Y_i = 1, T_i = 0 \mid Z_i = 0).\end{aligned}$$

- The width of this simple interval is

$$\begin{aligned}& \mathbb{P}(T_i = 0 \mid Z_i = 1) + \mathbb{P}(T_i = 1 \mid Z_i = 0) \\ & \quad = 1 - \{\mathbb{P}(T_i = 1 \mid Z_i = 1) - \mathbb{P}(T_i = 1 \mid Z_i = 0)\}.\end{aligned}$$

  With this orientation of Z_i, a larger first stage gives a narrower interval. Perfect compliance gives width zero.

Sharp Bounds for the ATE (1): Latent Types

- The simple IV bounds may leave information unused. **Sharp bounds** contain exactly the ATE values compatible with the observed data and our assumptions.
- Define a unit's latent type by

$$U_i = (T_i(0), T_i(1), Y_i(0), Y_i(1)) \in \{0,1\}^4.$$

  There are $2^4 = 16$ possible types.
- For a type $u = (t_0, t_1, y_0, y_1)$, write

$$t_z(u) = t_z, \qquad y_t(u) = y_t, \qquad p_u = \mathbb{P}(U_i = u).$$

  The type determines treatment under either instrument value and the outcome under either treatment.
- We know each possible type's treatment effect, $y_1(u) - y_0(u)$. The unknown quantities are the type shares p_u.

Sharp Bounds for the ATE (2): The Estimand

- By iterated expectations,

$$\begin{aligned}\tau &= \mathbb{E}[Y_i(1) - Y_i(0)] \\ &= \sum_u \mathbb{E}[Y_i(1) - Y_i(0) \mid U_i = u]\, p_u \\ &= \sum_u \{y_1(u) - y_0(u)\}\, p_u.\end{aligned}$$

  The last equality holds because U_i fixes both potential outcomes.

- Thus, the ATE is a linear function of the unknown type shares.

- Since the shares form a probability distribution,

$$p_u \geq 0 \quad \text{for every } u, \qquad \sum_u p_u = 1.$$

  We next use the observed data to constrain these shares.

Sharp Bounds for the ATE (3): Observed Data

- For every $y, t, z \in \{0, 1\}$, we observe $\mathbb{P}(Y_i = y, T_i = t \mid Z_i = z)$.
- Consistency, exclusion, and randomization give

$$\begin{aligned}
&\mathbb{P}(Y_i = y, T_i = t \mid Z_i = z) \\
&= \mathbb{P}(Y_i(t) = y, T_i(z) = t \mid Z_i = z) && \text{(Consistency and exclusion)} \\
&= \mathbb{P}(Y_i(t) = y, T_i(z) = t) && \text{(Randomization)} \\
&= \sum_u \mathbf{1}\{y_t(u) = y,\ t_z(u) = t\}\, p_u.
\end{aligned}$$

- The indicator selects the types that would produce observed cell (y, t) under instrument value z.
- One common distribution $\{p_u\}$ must reproduce all eight conditional cells, including both $Y_i = 0$ and $Y_i = 1$.

Sharp Bounds for the ATE (4): Optimization

- To obtain the lower and upper bounds, minimize and maximize the ATE over all type shares compatible with the data:

$$\min_{\{p_u\}} \Big/ \max \quad \sum_u \{y_1(u) - y_0(u)\}\, p_u$$

$$\text{subject to} \quad p_u \geq 0, \qquad \sum_u p_u = 1,$$

$$\mathbb{P}(Y_i = y, T_i = t \mid Z_i = z) = \sum_u \mathbf{1}\{y_t(u) = y,\ t_z(u) = t\}\, p_u \quad \text{for all } y, t, z.$$

- Every feasible distribution generates the observed data under our IV assumptions. The extrema therefore give sharp bounds.
- The no-IV Manski bounds are sharp under consistency alone. Stronger IV assumptions can yield a smaller interval.

Probability of Causation (1): Setup and Goal (Tian and Pearl 2000)

- Consider a population of patients. Let $T_i = 1$ denote receiving treatment and $T_i = 0$ denote receiving no treatment. Let $Y_i = 1$ denote survival and $Y_i = 0$ denote death.
- Each patient has two potential outcomes, $Y_i(1)$ and $Y_i(0)$. We maintain consistency:

$$Y_i = Y_i(T_i).$$

- Our goal is to learn the population share who would survive with treatment but die without it:

$$\text{Benefit} := \mathbb{P}(Y_i(0) = 0, Y_i(1) = 1),$$
$$\text{Harm} := \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0).$$

  Benefit is also called the probability of necessity and sufficiency.

- The ATE gives their difference:

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \text{Benefit} - \text{Harm}.$$

Probability of Causation (2): Data Sources

- We consider three cases, depending on the data available.
- **Observational data only**: we record treatment decisions and outcomes without randomizing treatment. We observe

$$\mathbb{P}(T_i = t, Y_i = y).$$

  Treatment may depend on unobserved health conditions.
- **Experimental data only**: we randomize treatment itself, with both arms having positive probability. Randomization and consistency identify

$$\mathbb{P}_{\text{exp}}(Y_i = 1 \mid T_i = t) = \mathbb{P}(Y_i(t) = 1).$$

  We learn the two potential-outcome marginals, but not their joint distribution.
- **Both sources**: we combine the observational joint distribution with the experimental potential-outcome marginals. Both sources must describe the same target population and potential outcomes.

Probability of Causation (3): Latent Probabilities

- Let T_i denote treatment under the population's observational assignment process. Define

$$p_{ty_0y_1} := \mathbb{P}(T_i = t, Y_i(0) = y_0, Y_i(1) = y_1).$$

  There are eight unknown probabilities.

- Split Benefit by observational treatment status:

$$\begin{aligned}\text{Benefit} &= \mathbb{P}(T_i = 1, Y_i(0) = 0, Y_i(1) = 1) \\ &\quad + \mathbb{P}(T_i = 0, Y_i(0) = 0, Y_i(1) = 1) \\ &= p_{101} + p_{001}.\end{aligned}$$

- The basic probability restrictions are

$$p_{ty_0y_1} \geq 0, \qquad \sum_{t=0}^{1}\sum_{y_0=0}^{1}\sum_{y_1=0}^{1} p_{ty_0y_1} = 1.$$

  We will add the restrictions supplied by the available data.

Probability of Causation (4): Observational Data

- First, suppose we observe the joint distribution $\mathbb{P}(T_i = t, Y_i = y)$ and maintain consistency, $Y_i = Y_i(T_i)$. Treatment may be confounded.
- For example, $T_i = 1, Y_i = 1$ reveals $Y_i(1) = 1$, while $Y_i(0)$ can be either 0 or 1. Therefore,

$$\mathbb{P}(T_i = 1, Y_i = 1) = p_{101} + p_{111}.$$

- Applying the same reasoning to all four cells gives

$$\begin{aligned}
\mathbb{P}(T_i = 1, Y_i = 1) &= p_{101} + p_{111}, \\
\mathbb{P}(T_i = 1, Y_i = 0) &= p_{100} + p_{110}, \\
\mathbb{P}(T_i = 0, Y_i = 1) &= p_{010} + p_{011}, \\
\mathbb{P}(T_i = 0, Y_i = 0) &= p_{000} + p_{001}.
\end{aligned}$$

We minimize and maximize $p_{101} + p_{001}$ subject to these equalities and the probability restrictions.

Probability of Causation (5): Observational Bounds

- Nonnegativity and the observed-cell restrictions imply

$$p_{101} \leq \mathbb{P}(T_i = 1, Y_i = 1), \qquad p_{001} \leq \mathbb{P}(T_i = 0, Y_i = 0).$$

- Since $\text{Benefit} = p_{101} + p_{001}$, we obtain

$$0 \leq \text{Benefit} \leq \mathbb{P}(T_i = 1, Y_i = 1) + \mathbb{P}(T_i = 0, Y_i = 0).$$

- Both endpoints are attainable:
  - For the upper bound, set $p_{111} = p_{000} = 0$. All mass in the two relevant observed cells can be Benefit.
  - For the lower bound, set $p_{101} = p_{001} = 0$. Neither observed cell needs to contain anyone who benefits.

  The remaining probabilities can match the other observed cells.

- Thus, these bounds are sharp under consistency alone.

Probability of Causation (6): Experimental Data

- Next, suppose treatment is randomized in an experiment. Consistency and randomization identify the population marginals:

$$\mathbb{P}(Y_i(t) = 1) = \mathbb{P}_{\text{exp}}(Y_i = 1 \mid T_i = t).$$

  Here $\mathbb{P}_{\text{exp}}$ denotes the experimental distribution, with both treatment arms having positive probability.

- Using the latent probabilities, these marginals impose

$$\mathbb{P}(Y_i(1) = 1) = p_{001} + p_{011} + p_{101} + p_{111},$$
$$\mathbb{P}(Y_i(0) = 1) = p_{010} + p_{011} + p_{110} + p_{111}.$$

  The probabilities for outcome 0 follow by subtraction from 1.

- We minimize and maximize $p_{101} + p_{001}$ subject to these marginal restrictions and the probability restrictions.

- The experiment identifies the ATE, but the joint distribution of $Y_i(0)$ and $Y_i(1)$ remains unknown.

Probability of Causation (7): Experimental Bounds

- Since $\tau = \text{Benefit} - \text{Harm}$ and $\text{Harm} \geq 0$,

$$\text{Benefit} \geq \max\{0, \tau\}.$$

- Benefit requires both $Y_i(1) = 1$ and $Y_i(0) = 0$. Thus,

$$\text{Benefit} \leq \min\{\mathbb{P}(Y_i(1) = 1), \mathbb{P}(Y_i(0) = 0)\}.$$

- To see sharpness, write the four potential-outcome shares as

$$\begin{aligned}
&\mathbb{P}(Y_i(0) = 0, Y_i(1) = 1) = \text{Benefit}, \\
&\mathbb{P}(Y_i(0) = 1, Y_i(1) = 0) = \text{Benefit} - \tau, \\
&\mathbb{P}(Y_i(0) = 1, Y_i(1) = 1) = \mathbb{P}(Y_i(1) = 1) - \text{Benefit}, \\
&\mathbb{P}(Y_i(0) = 0, Y_i(1) = 0) = \mathbb{P}(Y_i(0) = 0) - \text{Benefit}.
\end{aligned}$$

All four shares are nonnegative exactly when the bounds hold.

Probability of Causation (8): Combining Data

- Suppose we have observational data and experimental data. Both must describe the same target population and potential outcomes.
- The observational data identify

$$\mathbb{P}(T_i = t, Y_i = y),$$

  under the population's observational treatment assignment. The experiment identifies $\mathbb{P}(Y_i(1) = 1)$ and $\mathbb{P}(Y_i(0) = 1)$.
- We solve the same optimization problem for $\text{Benefit} = p_{101} + p_{001}$, now imposing *both* sets of data restrictions.
- The feasible set can only become smaller. Combining the sources may therefore tighten the bounds even though observational treatment is confounded.

Probability of Causation (9): Combined Lower Bound

- We already know that $\text{Benefit} \geq 0$ and $\text{Benefit} \geq \tau$.
- The two data sources give two further inequalities:

$$\mathbb{P}(Y_i = 1) - \mathbb{P}(Y_i(0) = 1) = p_{101} - p_{110} \leq \text{Benefit},$$
$$\mathbb{P}(Y_i(1) = 1) - \mathbb{P}(Y_i = 1) = p_{001} - p_{010} \leq \text{Benefit}.$$

  For example, $p_{101} - p_{110} \leq p_{101} \leq p_{101} + p_{001}$.

- Taking the largest of the four lower bounds gives

$$\text{Benefit} \geq \max \left\{ \begin{array}{l} 0, \\ \tau, \\ \mathbb{P}(Y_i = 1) - \mathbb{P}(Y_i(0) = 1), \\ \mathbb{P}(Y_i(1) = 1) - \mathbb{P}(Y_i = 1) \end{array} \right\}.$$

Probability of Causation (10): Combined Upper Bound

- We have three upper bounds from the separate data sources. A fourth follows from

$$\begin{aligned} \text{Harm} = p_{110} + p_{010} &\leq \mathbb{P}(T_i = 1, Y_i = 0) + \mathbb{P}(T_i = 0, Y_i = 1), \\ \text{Benefit} &= \tau + \text{Harm}. \end{aligned}$$

- Taking the smallest of the four upper bounds gives

$$\text{Benefit} \leq \min \left\{ \begin{array}{l} \mathbb{P}(Y_i(1) = 1), \\ \mathbb{P}(Y_i(0) = 0), \\ \mathbb{P}(T_i = 1, Y_i = 1) + \mathbb{P}(T_i = 0, Y_i = 0), \\ \tau + \mathbb{P}(T_i = 1, Y_i = 0) + \mathbb{P}(T_i = 0, Y_i = 1) \end{array} \right\}.$$

- The combined lower and upper bounds are sharp when the two data sources are compatible with one common population.

Probability of Causation (11): Monotonicity

- Suppose we additionally assume that treatment harms nobody:

$$Y_i(1) \geq Y_i(0) \quad \text{for every } i.$$

  This is monotonicity of the *outcome* in treatment.

- It rules out the harmful type:

$$\text{Harm} = \mathbb{P}(Y_i(0) = 1, Y_i(1) = 0) = p_{110} + p_{010} = 0.$$

- Consequently,

$$\begin{aligned} \text{Benefit} &= \tau + \text{Harm} \\ &= \mathbb{P}(Y_i(1) = 1) - \mathbb{P}(Y_i(0) = 1). \end{aligned}$$

  Experimental data therefore point identify Benefit.

- Observational data alone generally do not identify these marginals. IV treatment monotonicity, $T_i(1) \geq T_i(0)$, concerns a different relationship.

Regression Review: Setting

- Consider a full outcome model with a scalar confounder U_i:

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \delta U_i + \epsilon_i, \qquad \mathbb{E}[\epsilon_i \mid T_i, \boldsymbol{X}_i, U_i] = 0.$$

- Omitting U_i gives the population linear projection

$$Y_i = \alpha^* + \beta^* T_i + \boldsymbol{\gamma}^{*\top} \boldsymbol{X}_i + Y_i^{\perp T, \boldsymbol{X}}.$$

- Superscript $\perp$ denotes residuals from a linear projection including an intercept. For example, $T_i^{\perp \boldsymbol{X}}$ removes the linear association of T_i with $\boldsymbol{X}_i$.
- Assume finite second moments and positive residual variances. The omitted-variable bias is $\beta^* - \beta$.

Regression Review Question 1

- Derive the absolute omitted-variable bias formula:

$$|\beta^* - \beta| = \sqrt{R^2_{Y\sim U|T,\boldsymbol{X}} \frac{R^2_{T\sim U|\boldsymbol{X}}}{1 - R^2_{T\sim U|\boldsymbol{X}}} \frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}}.$$

- Start from the omitted-variable bias identity:

$$\beta^* - \beta = \delta \frac{\mathrm{Cov}(T_i^{\perp \boldsymbol{X}}, U_i^{\perp \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}.$$

- Hints: use the Frisch–Waugh–Lovell (FWL) theorem for δ and the fact that univariate R^2 equals squared correlation. The same relationship holds after residualizing both variables.

Regression Review Question 1 - Answer

- FWL gives the coefficient of U_i in the full model:

$$\delta = \frac{\text{Cov}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}.$$

- Multiply and divide by the outcome residual standard deviation:

$$\begin{aligned}|\delta| &= \frac{|\text{Cov}(Y_i^{\perp T,\boldsymbol{X}}, U_i^{\perp T,\boldsymbol{X}})|}{\sqrt{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}} \sqrt{\frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}} \\ &= \sqrt{R^2_{Y\sim U|T,\boldsymbol{X}} \frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}}.\end{aligned}$$

- The first factor is the absolute correlation between the residualized outcome and confounder.

Regression Review Question 1 - Answer (2)

- Apply the same idea to the other factor in the bias formula:

$$\begin{aligned}
&\frac{|\operatorname{Cov}(T_i^{\perp \boldsymbol{X}}, U_i^{\perp \boldsymbol{X}})|}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})} \\
&\quad = \frac{|\operatorname{Cov}(T_i^{\perp \boldsymbol{X}}, U_i^{\perp \boldsymbol{X}})|}{\sqrt{\mathbb{V}(T_i^{\perp \boldsymbol{X}})\mathbb{V}(U_i^{\perp \boldsymbol{X}})}} \sqrt{\frac{\mathbb{V}(U_i^{\perp \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}} \\
&\quad = \sqrt{R^2_{T \sim U|\boldsymbol{X}} \frac{\mathbb{V}(U_i^{\perp \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}}.
\end{aligned}$$

- Combining the two factors gives

$$|\beta^* - \beta| = \sqrt{R^2_{Y \sim U|T,\boldsymbol{X}} R^2_{T \sim U|\boldsymbol{X}} \frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})} \frac{\mathbb{V}(U_i^{\perp \boldsymbol{X}})}{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}}.$$

Regression Review Question 1 - Answer (3)

- The definition of partial R^2 gives

$$R^2_{U\sim T|\boldsymbol{X}} = 1 - \frac{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(U_i^{\perp \boldsymbol{X}})}.$$

- Because T_i and U_i are scalar, squared correlation is symmetric:

$$\begin{aligned} R^2_{U\sim T|\boldsymbol{X}} &= \mathrm{Cor}(U_i^{\perp \boldsymbol{X}}, T_i^{\perp \boldsymbol{X}})^2 \\ &= R^2_{T\sim U|\boldsymbol{X}}. \end{aligned}$$

- Therefore,

$$\frac{\mathbb{V}(U_i^{\perp \boldsymbol{X}})}{\mathbb{V}(U_i^{\perp T,\boldsymbol{X}})} = \frac{1}{1 - R^2_{T\sim U|\boldsymbol{X}}}.$$

  We require $R^2_{T\sim U|\boldsymbol{X}} < 1$ so that treatment still has residual variation after controlling for the confounder.

Regression Review Question 1 - Answer (4)

- Substitute the variance ratio into the previous expression:

$$|\beta^* - \beta| = \sqrt{R^2_{Y\sim U|T,\boldsymbol{X}} \frac{R^2_{T\sim U|\boldsymbol{X}}}{1 - R^2_{T\sim U|\boldsymbol{X}}} \frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}}.$$

- The first two terms measure the confounder's associations with the outcome and treatment, after adjusting for the stated variables.
- The last term converts these association measures to the scale of the treatment coefficient.

- The same algebra holds for sample OLS coefficients using empirical variances and partial R^2. If U_i is unobserved, its two partial R^2 values remain sensitivity parameters.

Regression Review: Standard Error Setting

- For Question 2, absorb the intercept into $\boldsymbol{X}_i$ and fit

$$Y_i = \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \epsilon_i.$$

  There are K non-intercept predictors: T_i and $K-1$ other predictors. Thus $\boldsymbol{Z} = [\boldsymbol{T}\ \boldsymbol{X}]$ has $K+1$ columns.

- Assume full column rank, $n > K+2$, and, for part (a),

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \boldsymbol{Z}] = \boldsymbol{0}, \qquad \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mid \boldsymbol{Z}] = \sigma^2 \boldsymbol{I}_n.$$

- In this question, $\mathbb{V}(A_i)$ for regression residuals denotes their **empirical variance**, with divisor $n-1$. The parameter σ^2 is the variance of the regression error.

Regression Review Question 2(a)

- Show that the conventional homoskedastic OLS standard error is

$$\widehat{\mathrm{se}}^*(\widehat{\beta}^*) = \sqrt{\frac{\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})} \frac{1}{n-K-1}}.$$

  Here $\widehat{\beta}^*$ is the coefficient from the model using only T_i and $\boldsymbol{X}_i$.

- Hint: use the upper-left entry of the block inverse:

$$\left[\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1}\right]_{11} = (A - BD^{-1}C)^{-1},$$

  where A is scalar. Apply this identity to $\boldsymbol{Z}^\top \boldsymbol{Z}$.

- Account for the $K+1$ estimated coefficients when estimating σ^2 from the residuals.

Regression Review Question 2(a) - Answer

- Write $\boldsymbol{\theta} = (\beta, \boldsymbol{\gamma}^\top)^\top$. The OLS estimator satisfies

$$\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{\epsilon}.$$

- Under the conditional mean and variance assumptions,

$$\begin{aligned}
\mathbb{V}(\widehat{\boldsymbol{\theta}} \mid \boldsymbol{Z}) &= (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mid \boldsymbol{Z}] \boldsymbol{Z} (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \\
&= \sigma^2 (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{Z} (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \\
&= \sigma^2 (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1}.
\end{aligned}$$

- Since the first column of $\boldsymbol{Z}$ is $\boldsymbol{T}$, the variance of $\widehat{\beta}^*$ is σ^2 times the upper-left entry of $(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1}$.

Regression Review Question 2(a) - Answer (2)

- Partition the cross-product matrix:

$$\boldsymbol{Z}^\top \boldsymbol{Z} = \begin{pmatrix} \boldsymbol{T}^\top \boldsymbol{T} & \boldsymbol{T}^\top \boldsymbol{X} \\ \boldsymbol{X}^\top \boldsymbol{T} & \boldsymbol{X}^\top \boldsymbol{X} \end{pmatrix}.$$

- The block-inverse formula gives

$$\begin{aligned} [(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1}]_{11} &= \left\{ \boldsymbol{T}^\top \boldsymbol{T} - \boldsymbol{T}^\top \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{T} \right\}^{-1} \\ &= \left\{ \boldsymbol{T}^\top \boldsymbol{M}_{\boldsymbol{X}} \boldsymbol{T} \right\}^{-1}, \end{aligned}$$

  where

$$\boldsymbol{P}_{\boldsymbol{X}} = \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top, \qquad \boldsymbol{M}_{\boldsymbol{X}} = \boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{X}}.$$

- $\boldsymbol{P}_{\boldsymbol{X}}$ gives fitted values from regression on $\boldsymbol{X}$; $\boldsymbol{M}_{\boldsymbol{X}}$ gives residuals.

Regression Review Question 2(a) - Answer (3)

- The residual-maker matrix is symmetric and idempotent:

$$\boldsymbol{M}_{\boldsymbol{X}}^{\top} = \boldsymbol{M}_{\boldsymbol{X}}, \qquad \boldsymbol{M}_{\boldsymbol{X}}^{2} = \boldsymbol{M}_{\boldsymbol{X}}.$$

- Hence the denominator is a sum of squared treatment residuals:

$$\begin{aligned} \boldsymbol{T}^{\top}\boldsymbol{M}_{\boldsymbol{X}}\boldsymbol{T} &= \boldsymbol{T}^{\top}\boldsymbol{M}_{\boldsymbol{X}}^{\top}\boldsymbol{M}_{\boldsymbol{X}}\boldsymbol{T} \\ &= (\boldsymbol{M}_{\boldsymbol{X}}\boldsymbol{T})^{\top}(\boldsymbol{M}_{\boldsymbol{X}}\boldsymbol{T}) \\ &= \sum_{i=1}^{n}(T_i^{\perp \boldsymbol{X}})^2 = (n-1)\mathbb{V}(T_i^{\perp \boldsymbol{X}}). \end{aligned}$$

- The last equality uses the intercept: residuals sum to zero. We have shown

$$\mathbb{V}(\widehat{\beta}^{*} \mid \boldsymbol{Z}) = \frac{\sigma^2}{(n-1)\mathbb{V}(T_i^{\perp \boldsymbol{X}})}.$$

Regression Review Question 2(a) - Answer (4)

- There are $K+1$ estimated coefficients. The unbiased estimator of the error variance is

$$\widehat{\sigma}^2 = \frac{\sum_{i=1}^{n}(Y_i^{\perp T, \boldsymbol{X}})^2}{n-K-1} = \frac{(n-1)\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}})}{n-K-1}.$$

- Substitute this estimate into the variance formula:

$$\begin{aligned}\widehat{\mathrm{se}}^*(\widehat{\beta}^*) &= \sqrt{\frac{\widehat{\sigma}^2}{(n-1)\mathbb{V}(T_i^{\perp \boldsymbol{X}})}} \\ &= \sqrt{\frac{\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}\frac{1}{n-K-1}}.\end{aligned}$$

- The common $n-1$ factor cancels. More residual outcome variation raises the standard error; more residual treatment variation lowers it.

Regression Review Question 2(b)

- Now suppose a correctly specified model also includes a scalar U_i:

$$Y_i = \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \delta U_i + \epsilon_i.$$

  Show that the full-model conventional standard error satisfies

$$\widehat{\mathrm{se}}(\widehat{\beta}) = \widehat{\mathrm{se}}^*(\widehat{\beta}^*)\sqrt{\frac{1 - R^2_{Y\sim U|T,\boldsymbol{X}}}{1 - R^2_{T\sim U|\boldsymbol{X}}}\frac{n-K-1}{n-K-2}}.$$

- Recall the empirical partial R^2 definition:

$$R^2_{Y\sim U|T,\boldsymbol{X}} = 1 - \frac{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X},U})}{\mathbb{V}(Y_i^{\perp T,\boldsymbol{X}})}.$$

- The full regression now estimates $K+2$ coefficients. Assume its errors are exogenous, uncorrelated, and homoskedastic.

Regression Review Question 2(b) - Answer

- Repeat part (a), adding U_i to the controls:

$$\widehat{\mathrm{se}}(\widehat{\beta}) = \sqrt{\frac{\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}, U})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}, U})} \frac{1}{n-K-2}}.$$

- Taking the ratio of the full and short conventional standard errors,

$$\frac{\widehat{\mathrm{se}}(\widehat{\beta})}{\widehat{\mathrm{se}}^*(\widehat{\beta}^*)} = \sqrt{\frac{\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}, U})}{\mathbb{V}(Y_i^{\perp T, \boldsymbol{X}})} \frac{\mathbb{V}(T_i^{\perp \boldsymbol{X}})}{\mathbb{V}(T_i^{\perp \boldsymbol{X}, U})} \frac{n-K-1}{n-K-2}}$$

$$= \sqrt{\frac{1-R^2_{Y \sim U | T, \boldsymbol{X}}}{1-R^2_{T \sim U | \boldsymbol{X}}} \frac{n-K-1}{n-K-2}}.$$

- The first two ratios become partial R^2 terms by definition. The final ratio accounts for estimating one additional coefficient.

Regression Review Question 2(b) - Answer (2)

- Adding U_i affects the standard error in three ways:
  1. It explains outcome variation left after $(T_i, \boldsymbol{X}_i)$: $1 - R^2_{Y \sim U | T, \boldsymbol{X}}$ lowers the residual variance.
  2. It explains treatment variation left after $\boldsymbol{X}_i$: $1 - R^2_{T \sim U | \boldsymbol{X}}$ lowers the denominator.
  3. It uses one degree of freedom: $n - K - 1$ becomes $n - K - 2$.

- Therefore, including a confounder can either increase or decrease the conventional standard error.

- The ratio is an algebraic identity for **conventional OLS** standard errors. Omitting U_i may violate the short model's homoskedasticity assumptions, so its reported standard error need not estimate its true sampling uncertainty correctly.

Regression Review: Selection Model Setting

- Consider the outcome and treatment equations:

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \epsilon_i,$$
$$T_i = \mathbf{1}\{\lambda + \boldsymbol{\delta}^\top \boldsymbol{X}_i + \eta_i > 0\}.$$

- Assume independent observations and conditional joint normality:

$$\begin{pmatrix} \epsilon_i \\ \eta_i \end{pmatrix} \Big| \, \boldsymbol{X}_i \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{pmatrix} \right].$$

- Let $a_i = \lambda + \boldsymbol{\delta}^\top \boldsymbol{X}_i$ and $W_i = 2T_i - 1$. Write ϕ and Φ for the standard normal density and CDF.
- Although $\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i] = 0$, treatment selection can give $\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i] \neq 0$ when $\rho \neq 0$.

Regression Review Question 3

(a) Show that

$$\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i] = \rho\sigma W_i \frac{\phi(\lambda + \boldsymbol{\delta}^\top \boldsymbol{X}_i)}{\Phi\{W_i(\lambda + \boldsymbol{\delta}^\top \boldsymbol{X}_i)\}} = \rho\sigma W_i \frac{\phi(a_i)}{\Phi(W_i a_i)}.$$

- Hint: first find $\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, \eta_i]$ using joint normality, then take expectations over the appropriate truncated normal distribution for η_i.

(b) Outline a two-step procedure for estimating the model: which regression do you fit first, and which additional predictor belongs in the second regression?

Regression Review Question 3(a) - Answer

- For a jointly normal pair, the conditional expectation is linear:

$$\begin{aligned}\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, \eta_i] &= \mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i] + \frac{\mathrm{Cov}(\epsilon_i, \eta_i \mid \boldsymbol{X}_i)}{\mathbb{V}(\eta_i \mid \boldsymbol{X}_i)}\{\eta_i - \mathbb{E}[\eta_i \mid \boldsymbol{X}_i]\} \\ &= \rho\sigma\eta_i.\end{aligned}$$

- Treatment is determined by $(\boldsymbol{X}_i, \eta_i)$. Thus iterated expectations give

$$\begin{aligned}\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i] &= \mathbb{E}[\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, \eta_i] \mid \boldsymbol{X}_i, T_i] \\ &= \rho\sigma\mathbb{E}[\eta_i \mid \boldsymbol{X}_i, T_i].\end{aligned}$$

- We now need the mean of η_i above $-a_i$ for treated units and below $-a_i$ for controls.

Regression Review Question 3(a) - Answer (2)

- If $T_i = 1$, then $\eta_i > -a_i$. The conditional density is

$$f(\eta \mid \boldsymbol{X}_i, T_i = 1) = \frac{\phi(\eta)\mathbf{1}\{\eta > -a_i\}}{1 - \Phi(-a_i)}.$$

- Since $\phi'(z) = -z\phi(z)$ and $\phi(\infty) = 0$,

$$\begin{aligned}\mathbb{E}[\eta_i \mid \boldsymbol{X}_i, T_i = 1] &= \frac{\int_{-a_i}^{\infty} \eta\phi(\eta)\, d\eta}{1 - \Phi(-a_i)} \\ &= \frac{[-\phi(\eta)]_{-a_i}^{\infty}}{1 - \Phi(-a_i)} = \frac{\phi(-a_i)}{1 - \Phi(-a_i)} \\ &= \frac{\phi(a_i)}{\Phi(a_i)}.\end{aligned}$$

- Therefore,

$$\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i = 1] = \rho\sigma\frac{\phi(a_i)}{\Phi(a_i)}.$$

Regression Review Question 3(a) - Answer (3)

- If $T_i = 0$, then $\eta_i \leq -a_i$. Similarly,

$$\mathbb{E}[\eta_i \mid \boldsymbol{X}_i, T_i = 0] = \frac{\int_{-\infty}^{-a_i} \eta \phi(\eta)\, d\eta}{\Phi(-a_i)}$$
$$= \frac{[-\phi(\eta)]_{-\infty}^{-a_i}}{\Phi(-a_i)} = -\frac{\phi(a_i)}{1 - \Phi(a_i)}.$$

- Combine the two cases using $W_i = 2T_i - 1$:

$$\mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i] = \rho\sigma W_i \frac{\phi(a_i)}{\Phi(W_i a_i)}.$$

- When $T_i = 1$, $W_i = 1$; when $T_i = 0$, $W_i = -1$ and $\Phi(-a_i) = 1 - \Phi(a_i)$, yielding the negative control-group term.

Regression Review Question 3(b) - Answer

- **Step 1: estimate treatment selection by probit.** Since

$$\Pr(T_i = 1 \mid \boldsymbol{X}_i) = \Phi(\lambda + \boldsymbol{\delta}^\top \boldsymbol{X}_i),$$

obtain $\widehat{\lambda}, \widehat{\boldsymbol{\delta}}$ by maximum likelihood and set

$$\widehat{a}_i = \widehat{\lambda} + \widehat{\boldsymbol{\delta}}^\top \boldsymbol{X}_i, \qquad \widehat{m}_i = W_i \frac{\phi(\widehat{a}_i)}{\Phi(W_i \widehat{a}_i)}.$$

- **Step 2: fit an outcome regression by OLS.** Include the estimated signed inverse Mills ratio:

$$Y_i = \alpha + \beta T_i + \boldsymbol{\gamma}^\top \boldsymbol{X}_i + \theta \widehat{m}_i + \text{error}, \qquad \theta = \rho\sigma.$$

- Both treatment groups contribute to this outcome regression, using the appropriate sign and denominator in $\widehat{m}_i$.

Regression Review Question 3(b) - Answer (2)

- Why does the correction work? At the true parameters, define

$$m_i = W_i \frac{\phi(a_i)}{\Phi(W_i a_i)}, \qquad v_i = \epsilon_i - \rho\sigma m_i.$$

  Then

$$\mathbb{E}[v_i \mid \boldsymbol{X}_i, T_i] = \mathbb{E}[\epsilon_i \mid \boldsymbol{X}_i, T_i] - \rho\sigma m_i = 0.$$

- Consistency requires the specified selection and outcome models, joint normality, and full rank in the second-stage regression. Without an excluded selection predictor, identification relies on the model's functional form.

- Inference must account for estimating $\widehat{m}_i$ and the second-stage heteroskedasticity, for example by bootstrapping **both steps** together.

Sensitivity Review: Setting

- LaLonde (1986) compares observational estimators with a randomized job-training experiment, the National Supported Work Demonstration (NSW).

- Compare two samples:
  1. **Experimental**: NSW treated and NSW randomized controls.
  2. **Observational**: the same NSW treated units and controls from the Panel Study of Income Dynamics (PSID).

- The quantity of interest is the average treatment effect for the treated:

$$\tau_{\text{ATT}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid T_i = 1].$$

  Can adjustment recover the experimental benchmark?

Sensitivity Review: Variables

- Use `nsw.csv` and `psid_controls.csv` from the observational-studies data directory.
- $T_i =$ `treat` indicates participation in job training; $Y_i =$ `re78` $= \log(1 +$ earnings in 1978$)$.
- The seven pre-treatment covariates $\boldsymbol{X}_i$ are
  - `age`, `education`;
  - `black`, `hispanic`, `married`, `nodegree`;
  - `re75`: $\log(1 +$ earnings in 1975$)$.
- Include treatment–covariate interactions to allow the treatment effect to vary with $\boldsymbol{X}_i$.

Sensitivity Review Question 1

- Estimate the ATT in the experimental and observational samples using
  1. the difference in means;
  2. linear regression;
  3. Heckman's two-step selection estimator.

- For the regression and selection estimators, adjust for all seven pre-treatment covariates and their interactions with treatment.

- Report point estimates and standard errors in a table. Briefly interpret the comparison with the experimental benchmark.

Sensitivity Review Question 1 - Answer (1)

- Let $n_t = \sum_i \mathbf{1}\{T_i = t\}$. The difference in means is

$$\widehat{\tau}_{\text{Diff}} = \frac{1}{n_1}\sum_i T_i Y_i - \frac{1}{n_0}\sum_i (1 - T_i) Y_i.$$

- With independent observations, its estimated standard error is

$$\widehat{\text{se}}(\widehat{\tau}_{\text{Diff}}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}, \qquad s_t^2 = \frac{\sum_{i:T_i=t}(Y_i - \overline{Y}_t)^2}{n_t - 1}.$$

- Random assignment identifies the treatment effect in the experiment. With PSID controls, the raw difference can also reflect selection bias.

Sensitivity Review Question 1 - Answer (2)

- Fit the interacted outcome regression:

$$\mu_t(\boldsymbol{x}) = \alpha + \beta t + \boldsymbol{x}^\top \boldsymbol{\gamma} + t\boldsymbol{x}^\top \boldsymbol{\delta}.$$

- Predict untreated outcomes for the treated and average:

$$\widehat{\tau}_{\text{Reg}} = \frac{1}{n_1} \sum_{i:T_i=1} \{Y_i - \widehat{\mu}_0(\boldsymbol{X}_i)\}.$$

- OLS residuals sum to zero within the treated group, so equivalently

$$\widehat{\tau}_{\text{Reg}} = \widehat{\beta} + \overline{\boldsymbol{X}}_1^\top \widehat{\boldsymbol{\delta}}, \qquad \overline{\boldsymbol{X}}_1 = \frac{1}{n_1} \sum_{i:T_i=1} \boldsymbol{X}_i.$$

- For a causal ATT, we need consistency, overlap for the treated, unconfoundedness, and a correct untreated-outcome model.

Sensitivity Review Question 1 - Answer (3)

- For Heckman's estimator, first fit a probit for treatment and form

$$\widehat{a}_i = \widehat{\lambda} + \boldsymbol{X}_i^\top \widehat{\boldsymbol{\pi}}, \qquad \widehat{m}_i = (2T_i - 1)\frac{\phi(\widehat{a}_i)}{\Phi((2T_i - 1)\widehat{a}_i)}.$$

- Then estimate the selection-adjusted outcome regression:

$$Y_i = \alpha + \beta T_i + \boldsymbol{X}_i^\top \boldsymbol{\gamma} + T_i \boldsymbol{X}_i^\top \boldsymbol{\delta} + \theta \widehat{m}_i + v_i.$$

- Under the common-error selection model, both potential outcomes for a treated unit condition on **the same treated selection event**:

$$\widehat{\mathbb{E}}[Y_i(t) \mid \boldsymbol{X}_i, T_i = 1] = \widehat{\alpha} + \widehat{\beta} t + \boldsymbol{X}_i^\top \widehat{\boldsymbol{\gamma}} + t \boldsymbol{X}_i^\top \widehat{\boldsymbol{\delta}} + \widehat{\theta} \frac{\phi(\widehat{a}_i)}{\Phi(\widehat{a}_i)}.$$

Sensitivity Review Question 1 - Answer (4)

- Subtract the two potential-outcome predictions for a treated unit: the common selection correction cancels. Thus,

$$\widehat{\tau}_{\text{Heckman}} = \widehat{\beta} + \overline{\boldsymbol{X}}_1^{\top} \widehat{\boldsymbol{\delta}}.$$

  The coefficients come from the selection-adjusted regression.

- When predicting $Y_i(0)$ for treated units, keep their treated inverse Mills ratio. Substituting the control-group ratio would compare different selection groups and would not estimate the ATT.

- For regression and Heckman standard errors, bootstrap observations 1,000 times and refit the entire estimator in each draw. For Heckman, this includes both the probit and outcome stages.

Sensitivity Review Question 1 - Answer (5)

- Estimated effects on log(1 + earnings) (SEs in parentheses):

Estimator	NSW experiment	NSW treated + PSID
Difference in means	0.669 (0.290)	−2.160 (0.224)
Regression	0.608 (0.288)	0.410 (0.319)
Heckman two-step	198.552† (122.876)	1.708 (0.636)

- The observational sample retains all NSW treated units and replaces the randomized controls with PSID controls.
- † The experimental Heckman estimate is numerically unstable; see the next slide. Both-stage bootstrap SEs use seed 225.

Sensitivity Review Question 1 - Answer (6)

- Why is the experimental Heckman estimate so large? The estimated ATT is 198.552 with bootstrap SE 122.876.
- With nearly random treatment assignment, fitted selection probabilities vary little with covariates. The inverse Mills ratio is nearly collinear with the outcome regressors.
- Without an excluded selection predictor, separating the treatment effect from the selection correction is unstable. With a truly constant propensity, the correction is collinear with the intercept and treatment.
- The experimental difference in means provides a direct benchmark. A more elaborate selection model can be much less reliable here.

Sensitivity Review Question 1 - Answer (7)

- The experimental difference and regression estimates are about 0.669 and 0.608. With PSID controls, the estimates are -2.160 (raw difference), 0.410 (regression), and 1.708 (selection model).
- Regression adjustment changes the raw observational comparison, but its validity still depends on overlap and unconfoundedness.
- Heckman's estimator replaces unconfoundedness with a parametric selection model. Here there is no excluded selection predictor, so identification relies on functional form and joint normality.
- Here $Y_i = \log(1 + \text{earnings})$, so $\exp(\widehat{\tau})$ describes a ratio of geometric means of $1 + \text{earnings}$. It is not generally a ratio of arithmetic mean earnings.

Sensitivity Review Question 2

- Use the regression fitted to the NSW-treated/PSID-control sample in Question 1.
- Create a contour plot of the absolute omitted-variable bias as a function of the two partial R^2 values:
  - the association between U_i and treatment;
  - the association between U_i and outcome, given treatment.
- Use the bias formula from Regression Review Question 1 and briefly interpret the plot.
- With treatment interactions, first express the ATT as a single regression coefficient, as on the next slide.

Sensitivity Review Question 2 - Answer (1)

- Center covariates at the treated mean: $\boldsymbol{C}_i = \boldsymbol{X}_i - \overline{\boldsymbol{X}}_1$. Refit the same regression as

$$Y_i = \alpha_c + \tau T_i + \boldsymbol{C}_i^\top \boldsymbol{\gamma}_c + T_i \boldsymbol{C}_i^\top \boldsymbol{\delta} + e_i.$$

  This only changes the parameterization: $\widehat{\tau} = \widehat{\beta} + \overline{\boldsymbol{X}}_1^\top \widehat{\boldsymbol{\delta}} = \widehat{\tau}_{\mathrm{Reg}}$.

- Let $\boldsymbol{Z}_i = (1, \boldsymbol{C}_i^\top, T_i \boldsymbol{C}_i^\top)^\top$ collect all nuisance regressors. Define

$$r_T = R^2_{T \sim U | \boldsymbol{Z}}, \qquad r_Y = R^2_{Y \sim U | T, \boldsymbol{Z}}.$$

- Residualize on **all** nuisance columns, including interactions. This makes the coefficient bias formula apply to the ATT coefficient. Residualizing treatment only on baseline covariates would target a different coefficient.

Sensitivity Review Question 2 - Answer (2)

- Use sample residual variances from the centered regression:

$$S = \frac{\widehat{\mathbb{V}}(Y_i^{\perp T, Z})}{\widehat{\mathbb{V}}(T_i^{\perp Z})}.$$

- Regression Review Question 1 gives the absolute bias:

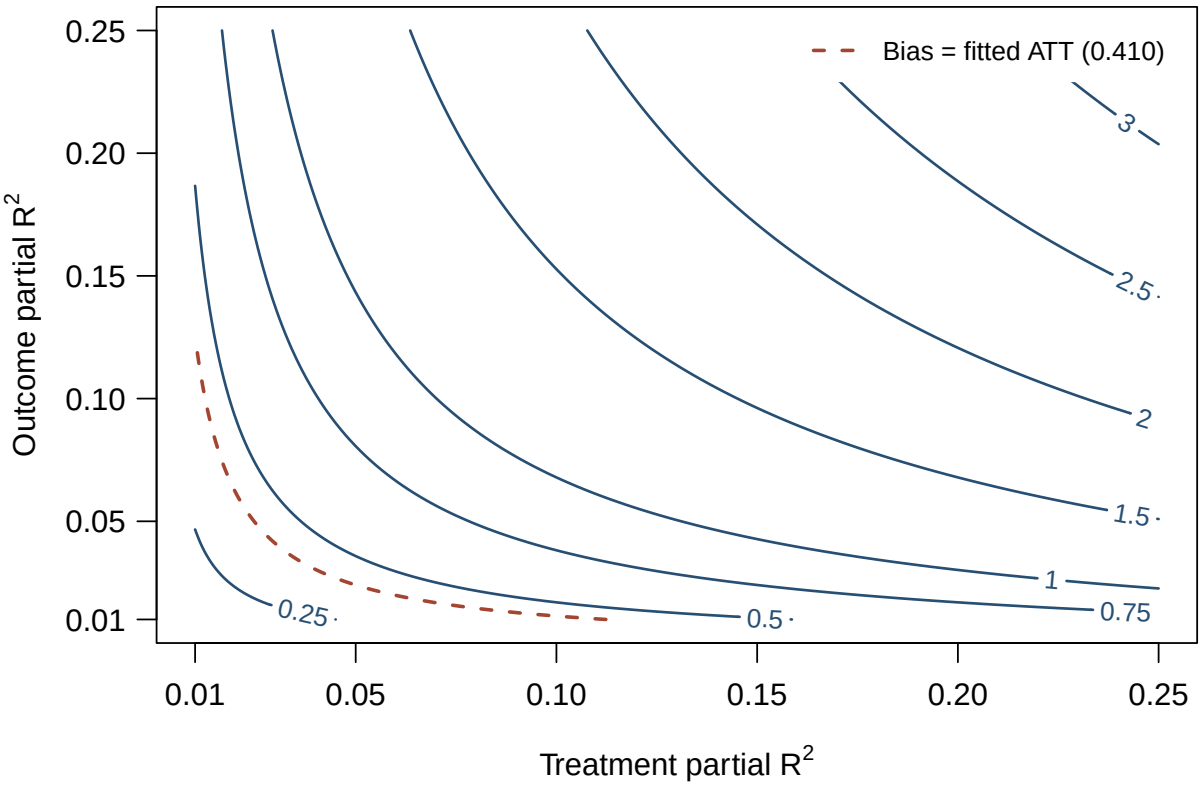
$$B(r_T, r_Y) = \sqrt{r_Y \frac{r_T}{1 - r_T} S}.$$

- Here $S \approx 132.573$ and $\widehat{\tau}_{\mathrm{Reg}} \approx 0.410$. At $r_T = r_Y = 0.05$,

$$B \approx 0.591.$$

A confounder explaining 5% of both residual variations could produce bias larger than the fitted ATT coefficient.

Sensitivity Review Question 2 - Answer (3)



- Bias rises as either partial R^2 increases. Above the dashed contour, positive bias can explain away the fitted positive effect.
- The axes describe hypothetical confounding strengths; these strengths cannot be estimated from the observed data alone.

Sensitivity Review Question 3

- Assume the omitted-variable bias is **positive**: the short-regression coefficient overstates the adjusted coefficient.
- Using the bias and standard-error results from the regression review, test

$$H_0 : \tau_{\mathrm{ATT}} = 0$$

  for different values of (r_T, r_Y).

- Plot the regions in which a two-sided test rejects at the 5% level. Distinguish positive and negative adjusted effects.
- Use the centered coefficient from Question 2, treating the observed design and centering point as fixed for conventional OLS inference.

Sensitivity Review Question 3 - Answer (1)

- There are $n = 2{,}787$ observations and $K = 15$ slopes (treatment, seven covariates, and seven interactions). Thus,

$$d = n - K - 1 = 2{,}771.$$

  The short-regression coefficient and conventional standard error are $\widehat{\tau}^* = 0.4095$ and $\widehat{\mathrm{se}}^* = 0.2187$.

- For one scalar omitted confounder, adjust both the coefficient and SE:

$$\widehat{\tau}(r_T, r_Y) = \widehat{\tau}^* - B(r_T, r_Y),$$
$$\widehat{\mathrm{se}}(r_T, r_Y) = \widehat{\mathrm{se}}^* \sqrt{\frac{1 - r_Y}{1 - r_T} \frac{d}{d-1}}.$$

- This SE relation uses homoskedastic OLS with the fixed design. The bootstrap SE from Question 1 is not substituted into this identity.

Sensitivity Review Question 3 - Answer (2)

- For each pair of sensitivity parameters, calculate

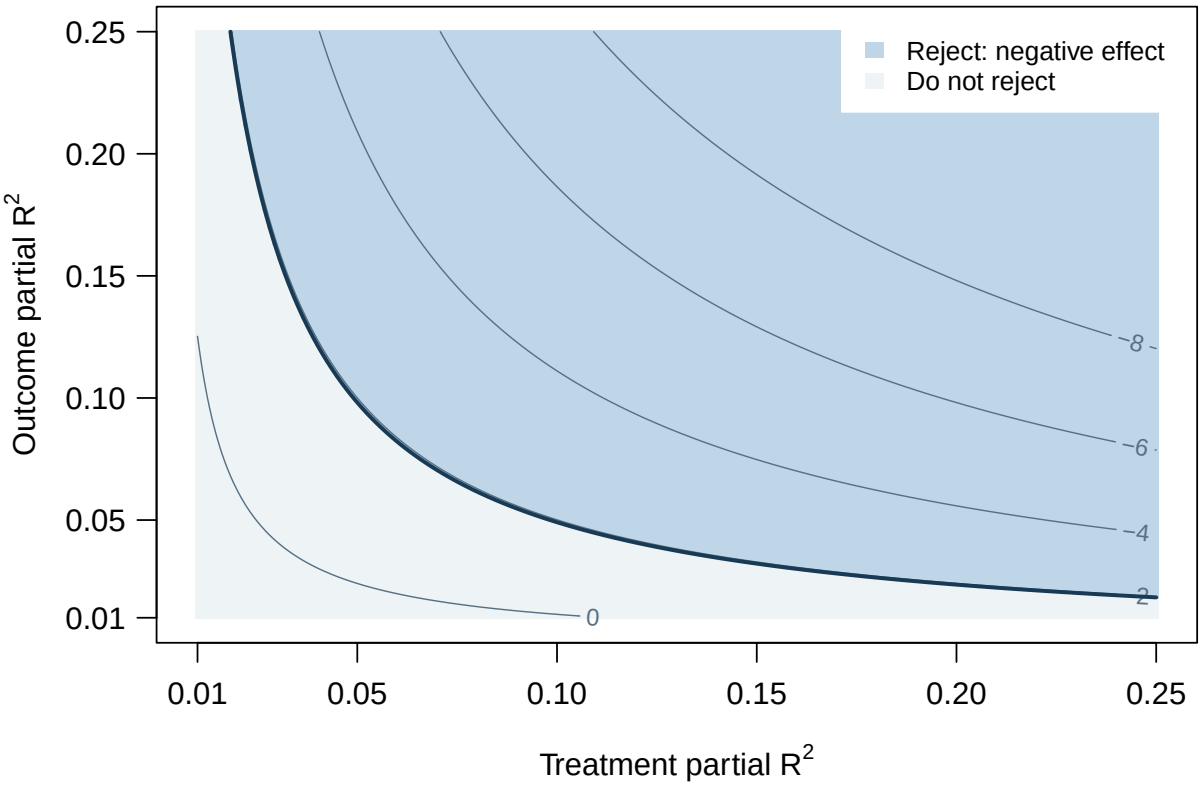
$$t(r_T, r_Y) = \frac{\widehat{\tau}^* - B(r_T, r_Y)}{\widehat{\mathrm{se}}^* \sqrt{\frac{1-r_Y}{1-r_T}\frac{d}{d-1}}}.$$

- Using the conventional t reference, reject at 5% if

$$|t(r_T, r_Y)| > t_{0.975,d-1} \approx 1.961.$$

- At $r_T = r_Y = 0.05$, the adjusted coefficient is about -0.181 and $t \approx -0.83$: we do not reject.
- At $r_T = r_Y = 0.10$, it is about -0.804 and $t \approx -3.68$: the test rejects with a negative adjusted effect.

Sensitivity Review Question 3 - Answer (3)



- Sufficient positive omitted-variable bias can reverse the estimated sign and produce rejection in the negative direction.
- This conclusion is conditional on the specified bias direction, confounding strengths, and outcome-model assumptions.

Appendix: Heckman's Selection Model (1)

- Consider the outcome and treatment models:

$$Y_i = \alpha + \beta T_i + X_i^\top \gamma + \epsilon_i,$$
$$T_i = \mathbf{1}\{\lambda + X_i^\top \delta + \eta_i > 0\},$$

- Assume joint normality conditional on X_i:

$$\begin{bmatrix} \epsilon_i \\ \eta_i \end{bmatrix} \Big| X_i \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{bmatrix} \right).$$

- Thus $\mathbb{E}[\epsilon_i \mid X_i] = 0$ (i.e., X_i is exogenous), but selection on η_i can make $\mathbb{E}[\epsilon_i \mid X_i, T_i] \neq 0$ when $\rho \neq 0$ (i.e., T_i is endogeneous).

Appendix: Heckman's Selection Model (2)

- Let ϕ and Φ be the standard normal density and CDF. The treatment model implies

$$\begin{aligned}\Pr(T_i = 1 \mid X_i) &= \Pr(\eta_i > -(\lambda + X_i^\top \delta) \mid X_i) \\ &= 1 - \Phi(-\lambda - X_i^\top \delta) = \Phi(\lambda + X_i^\top \delta).\end{aligned}$$

  Thus (λ, δ) can be estimated by probit.

- Project ϵ_i onto η_i, conditional on X_i:

$$\epsilon_i = \underbrace{\frac{\mathrm{Cov}(\epsilon_i, \eta_i \mid X_i)}{\mathbb{V}(\eta_i \mid X_i)}}_{=\rho\sigma/1} \eta_i + \widetilde{\epsilon}_i.$$

$$\mathrm{Cov}(\widetilde{\epsilon}_i, \eta_i \mid X_i) = \rho\sigma - \rho\sigma\mathbb{V}(\eta_i \mid X_i) = 0.$$

  Joint normality turns zero covariance into conditional independence: $\mathbb{E}[\widetilde{\epsilon}_i \mid X_i, \eta_i] = \mathbb{E}[\widetilde{\epsilon}_i \mid X_i] = 0$.

Appendix: Heckman's Selection Model (3)

- Because (X_i, η_i) determines T_i, iterated expectations give

$$\begin{aligned}\mathbb{E}[\widetilde{\epsilon}_i \mid X_i, T_i] &= \mathbb{E}[\mathbb{E}[\widetilde{\epsilon}_i \mid X_i, \eta_i] \mid X_i, T_i] \\ &= \mathbb{E}[0 \mid X_i, T_i] = 0.\end{aligned}$$

- Taking conditional expectations of $\epsilon_i = \rho\sigma\eta_i + \widetilde{\epsilon}_i$, we obtain

$$\begin{aligned}\mathbb{E}[\epsilon_i \mid X_i, T_i] &= \rho\sigma\mathbb{E}[\eta_i \mid X_i, T_i] + \mathbb{E}[\widetilde{\epsilon}_i \mid X_i, T_i] \\ &= \rho\sigma\mathbb{E}[\eta_i \mid X_i, T_i].\end{aligned}$$

- We therefore need the mean of a standard normal variable conditional on being above or below $-(\lambda + X_i^\top \delta)$.

Appendix: Truncated Normal Distribution

- We use the following facts about the truncated normal distribution
- For $Z \sim \mathcal{N}(0,1)$, we know that

$$\phi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}, \qquad \phi'(z) = -z\phi(z), \qquad \phi(\pm\infty) = 0.$$

- Conditioning on $Z > c$ divides $\phi(z)$ by its probability $1 - \Phi(c)$:

$$\begin{aligned}\mathbb{E}[Z \mid Z > c] &= \int_c^\infty z \, \frac{\phi(z)}{1-\Phi(c)} \, dz \\ &= \frac{-\int_c^\infty \phi'(z)\, dz}{1-\Phi(c)} = \frac{-\{0 - \phi(c)\}}{1-\Phi(c)} = \frac{\phi(c)}{1-\Phi(c)}.\end{aligned}$$

- For the lower tail, the same calculation gives

$$\begin{aligned}\mathbb{E}[Z \mid Z \le c] &= \int_{-\infty}^c z \, \frac{\phi(z)}{\Phi(c)} \, dz \\ &= \frac{-\int_{-\infty}^c \phi'(z)\, dz}{\Phi(c)} = \frac{-\{\phi(c) - 0\}}{\Phi(c)} = -\frac{\phi(c)}{\Phi(c)}.\end{aligned}$$

Appendix: Heckman's Selection Model (4)

- Treated units satisfy $\eta_i > -(\lambda + X_i^\top \delta)$. Set $c = -(\lambda + X_i^\top \delta)$ in the upper-tail formula:

$$\begin{aligned}\mathbb{E}[\eta_i \mid X_i, T_i = 1] &= \mathbb{E}[\eta_i \mid X_i, \eta_i > -(\lambda + X_i^\top \delta)] \\ &= \frac{\phi(-(\lambda + X_i^\top \delta))}{1 - \Phi(-(\lambda + X_i^\top \delta))} = \frac{\phi(\lambda + X_i^\top \delta)}{\Phi(\lambda + X_i^\top \delta)}.\end{aligned}$$

- Controls satisfy $\eta_i \leq -(\lambda + X_i^\top \delta)$. The lower-tail formula gives

$$\begin{aligned}\mathbb{E}[\eta_i \mid X_i, T_i = 0] &= \mathbb{E}[\eta_i \mid X_i, \eta_i \leq -(\lambda + X_i^\top \delta)] \\ &= -\frac{\phi(-(\lambda + X_i^\top \delta))}{\Phi(-(\lambda + X_i^\top \delta))} = -\frac{\phi(\lambda + X_i^\top \delta)}{1 - \Phi(\lambda + X_i^\top \delta)}.\end{aligned}$$

- Both simplifications use normal symmetry: $\phi(-(\lambda + X_i^\top \delta)) = \phi(\lambda + X_i^\top \delta)$ and $\Phi(-(\lambda + X_i^\top \delta)) = 1 - \Phi(\lambda + X_i^\top \delta)$.

Appendix: Heckman's Selection Model (5)

- Combining the two groups gives

$$m_i := \mathbb{E}[\eta_i \mid X_i, T_i] = T_i \frac{\phi(\lambda + X_i^\top \delta)}{\Phi(\lambda + X_i^\top \delta)} - (1 - T_i) \frac{\phi(\lambda + X_i^\top \delta)}{1 - \Phi(\lambda + X_i^\top \delta)}$$
$$= (2T_i - 1) \frac{\phi(\lambda + X_i^\top \delta)}{\Phi((2T_i - 1)(\lambda + X_i^\top \delta))}.$$

  which is called the inverse Mills ratio.

- Let $\theta = \rho\sigma$ and $v_i = \epsilon_i - \theta m_i$. Then consider

$$Y_i = \alpha + \beta T_i + X_i^\top \gamma + \theta m_i + v_i$$

  Notice that the selection bias is solved by including m_i in the outcome regression since

$$\mathbb{E}[v_i \mid X_i, T_i] = \mathbb{E}[\epsilon_i \mid X_i, T_i] - \rho\sigma m_i = \rho\sigma m_i - \rho\sigma m_i = 0.$$

- Estimate $\lambda + X_i^\top \delta$ by probit, construct $\widehat{m}_i$, and include it in the outcome regression.